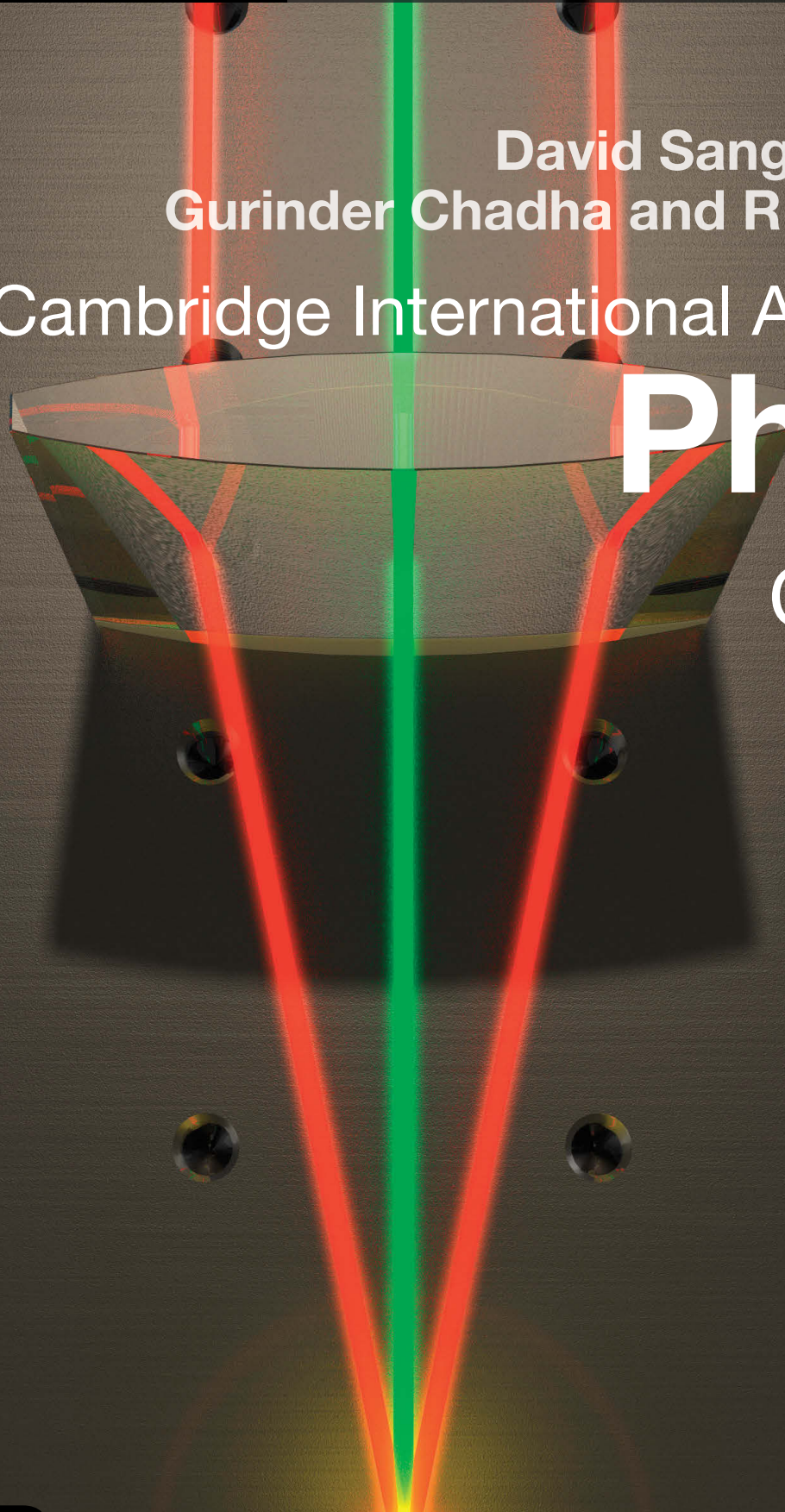




David Sang, Graham Jones,
Gurinder Chadha and Richard Woodside

Cambridge International AS and A Level

Physics

Coursebook

Second Edition



Completely Cambridge
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**David Sang, Graham Jones,
Gurinder Chadha and Richard Woodside**

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Example answers and all other end-of-chapter questions were written by the authors.

Contents

Introduction	vii	Chapter 5: Work, energy and power	69
How to use this book	viii	Doing work, transferring energy	71
Chapter 1: Kinematics – describing motion	1	Gravitational potential energy	75
Speed	2	Kinetic energy	76
Distance and displacement, scalar and vector	4	g.p.e.–k.e. transformations	76
Speed and velocity	5	Down, up, down – energy changes	77
Displacement–time graphs	6	Energy transfers	78
Combining displacements	8	Power	80
Combining velocities	10	Chapter 6: Momentum	85
Chapter 2: Accelerated motion	14	The idea of momentum	86
The meaning of acceleration	15	Modelling collisions	86
Calculating acceleration	15	Understanding collisions	89
Units of acceleration	16	Explosions and crash-landings	91
Deducing acceleration	17	Collisions in two dimensions	93
Deducing displacement	17	Momentum and Newton’s laws	95
Measuring velocity and acceleration	18	Understanding motion	96
Determining velocity and acceleration in the laboratory	18	Chapter 7: Matter and materials	101
The equations of motion	20	Density	102
Deriving the equations of motion	22	Pressure	102
Uniform and non-uniform acceleration	24	Compressive and tensile forces	104
Acceleration caused by gravity	25	Stretching materials	105
Determining g	25	Elastic potential energy	108
Motion in two dimensions – projectiles	28	Chapter 8: Electric fields	116
Understanding projectiles	29	Attraction and repulsion	117
Chapter 3: Dynamics – explaining motion	37	The concept of an electric field	118
Calculating the acceleration	38	Electric field strength	119
Understanding SI units	39	Force on a charge	122
The pull of gravity	41	Chapter 9: Electric current, potential difference and resistance	127
Mass and inertia	43	Circuit symbols and diagrams	128
Top speed	44	Electric current	129
Moving through fluids	45	An equation for current	132
Identifying forces	47	The meaning of voltage	134
Newton’s third law of motion	49	Electrical resistance	135
Chapter 4: Forces – vectors and moments	53	Electrical power	136
Combining forces	54	Chapter 10: Kirchhoff’s laws	143
Components of vectors	56	Kirchhoff’s first law	144
Centre of gravity	59	Kirchhoff’s second law	145
The turning effect of a force	59	Applying Kirchhoff’s laws	146
The torque of a couple	63	Resistor combinations	148

Chapter 11: Resistance and resistivity 156

The I - V characteristic for a metallic conductor	157
Ohm's law	158
Resistance and temperature	159
Resistivity	162

Chapter 12: Practical circuits 168

Internal resistance	169
Potential dividers	172
Potentiometer circuits	172

Chapter 13: Waves 178

Describing waves	179
Longitudinal and transverse waves	181
Wave energy	182
Wave speed	183
The Doppler effect	184
Electromagnetic waves	185
Electromagnetic radiation	186
Orders of magnitude	187
The nature of electromagnetic waves	188

Chapter 14: Superposition of waves 192

The principle of superposition of waves	193
Diffraction of waves	194
Interference	196
The Young double-slit experiment	200
Diffraction gratings	203

Chapter 15: Stationary waves 210

From moving to stationary	211
Nodes and antinodes	212
Formation of stationary waves	212
Determining the wavelength and speed of sound	216

Chapter 16: Radioactivity 222

Looking inside the atom	223
Alpha-particle scattering and the nucleus	223
A simple model of the atom	225
Nucleons and electrons	226
Forces in the nucleus	229
Fundamental particles?	229
Families of particles	230
Discovering radioactivity	231
Radiation from radioactive substances	231
Discovering neutrinos	232
Fundamental families	232
Fundamental forces	232
Properties of ionising radiation	233

P1: Practical skills at AS level 239

Practical work in physics	240
Using apparatus and following instructions	240
Gathering evidence	241
Precision, accuracy, errors and uncertainties	241
Finding the value of an uncertainty	243
Percentage uncertainty	245
Recording results	246
Analysing results	246
Testing a relationship	248
Identifying limitations in procedures and suggesting improvements	250

Chapter 17: Circular motion 258

Describing circular motion	259
Angles in radians	260
Steady speed, changing velocity	261
Angular velocity	261
Centripetal forces	262
Calculating acceleration and force	264
The origins of centripetal forces	265

Chapter 18: Gravitational fields 271

Representing a gravitational field	272
Gravitational field strength g	274
Energy in a gravitational field	276
Gravitational potential	276
Orbiting under gravity	277
The orbital period	278
Orbiting the Earth	279

Chapter 19: Oscillations 285

Free and forced oscillations	286
Observing oscillations	287
Describing oscillations	288
Simple harmonic motion	289
Representing s.h.m. graphically	291
Frequency and angular frequency	292
Equations of s.h.m.	293
Energy changes in s.h.m.	296
Damped oscillations	297
Resonance	299

Chapter 20: Communications systems 309

Radio waves	310
Analogue and digital signals	314
Channels of communication	317
Comparison of different channels	319

Chapter 21: Thermal physics	327	Chapter 27: Charged particles	422
Changes of state	328	Observing the force	423
Energy changes	329	Orbiting charges	423
Internal energy	331	Electric and magnetic fields	427
The meaning of temperature	332	The Hall effect	428
Thermometers	334	Discovering the electron	429
Calculating energy changes	336		
Chapter 22: Ideal gases	345	Chapter 28: Electromagnetic induction	435
Particles of a gas	346	Observing induction	436
Explaining pressure	348	Explaining electromagnetic induction	437
Measuring gases	348	Faraday's law of electromagnetic induction	441
Boyle's law	349	Lenz's law	443
Changing temperature	350	Using induction: eddy currents, generators and transformers	445
Ideal gas equation	351		
Modelling gases – the kinetic model	352	Chapter 29: Alternating currents	451
Temperature and molecular kinetic energy	354	Sinusoidal current	452
		Alternating voltages	453
Chapter 23: Coulomb's law	359	Power and a.c.	455
Electric fields	360	Why use a.c. for electricity supply?	457
Coulomb's law	360	Transformers	458
Electric field strength for a radial field	362	Rectification	460
Electric potential	363		
Comparing gravitational and electric fields	366	Chapter 30: Quantum physics	466
		Modelling with particles and waves	467
Chapter 24: Capacitance	372	Particulate nature of light	468
Capacitors in use	373	The photoelectric effect	471
Energy stored in a capacitor	375	Line spectra	475
Capacitors in parallel	377	Explaining the origin of line spectra	476
Capacitors in series	378	Photon energies	477
Comparing capacitors and resistors	379	Electron energies in solids	478
Capacitor networks	380	The nature of light – waves or particles?	480
		Electron waves	480
Chapter 25: Electronics	386		
Components of an electronic sensing system	387	Chapter 31: Nuclear physics	489
The operational amplifier (op-amp)	393	Balanced equations	490
The inverting amplifier	397	Mass and energy	491
The non-inverting amplifier	398	Energy released in radioactive decay	494
Output devices	398	Binding energy and stability	494
		Randomness and decay	496
Chapter 26: Magnetic fields and electromagnetism	406	The mathematics of radioactive decay	497
Producing and representing magnetic fields	407	Decay graphs and equations	499
Magnetic force	409	Decay constant and half-life	501
Magnetic flux density	411		
Measuring magnetic flux density	411		
Currents crossing fields	413		
Forces between currents	415		
Relating SI units	416		
Comparing forces in magnetic, electric and gravitational fields	417		

Chapter 32: Medical imaging 506

The nature and production of X-rays	507
X-ray attenuation	509
Improving X-ray images	511
Computerised axial tomography	513
Using ultrasound in medicine	516
Echo sounding	518
Ultrasound scanning	520
Magnetic resonance imaging	522

P2: Planning, analysis and evaluation 529

Planning	530
Analysis of the data	532
Treatment of uncertainties	536
Conclusions and evaluation of results	538

Appendix 1: Physical quantities and units 542

Prefixes	542
Estimation	542

Appendix 2: Data, formulae and relationships 543

Data	543
Conversion factors	543
Mathematical equations	544
Formulae and relationships	544

Appendix 3: The Periodic Table 545**Glossary 546****Index 555****Acknowledgements 564****CD-ROM CD1**

Advice on how to revise for and approach examinations	CD1
Introduction to the examination and changes to the syllabus	CD9
Answers to self-assessment questions	CD12
Answers to end-of-chapter questions	CD70
Recommended resources	CD136

Introduction

This book covers the entire syllabus of Cambridge International Examinations AS and A Level Physics. It is designed to work with the syllabus that will be examined from 2016. It is in three parts:

- **Chapters 1–16 and P1:** the AS level content, covered in the first year of the course, including a **chapter (P1)** dedicated to the development of your practical skills
- **Chapters 17–32 and P2:** the remaining A level content, including a **chapter (P2)** dedicated to developing your ability to plan, analyse and evaluate practical investigations
- **Appendices** of useful formulae, a Glossary and an Index.

The main tasks of a textbook like this are to explain the various concepts of physics that you need to understand and to provide you with questions that will help you to test your understanding and prepare for your examinations. You will find a visual guide to the structure of each chapter and the features of this book on the next two pages.

When tackling questions, it is a good idea to make a first attempt without referring to the explanations in this Coursebook or to your notes. This will help to reveal any gaps in your understanding. By working out which concepts you find most challenging, and by spending more time to understand these concepts at an early stage, you will progress faster as the course continues.

The CD-ROM that accompanies this Coursebook includes answers with workings for all the questions in the book, as well as suggestions for revising and preparing for any examinations you take. There are also lists of recommended further reading, which in many cases will take you beyond the requirements of the syllabus, but which will help you deepen your knowledge and explain more of the background to the physics concepts covered in this Coursebook.

In your studies, you will find that certain key concepts come up again and again, and that these concepts form ‘themes’ that link the different areas of physics together. It will help you to progress and gain confidence in tackling problems if you take note of these themes. For this Coursebook, these key concepts include:

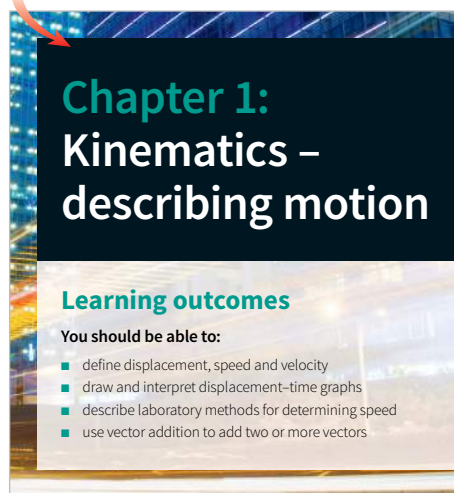
- Models of physical systems
- Testing predictions against evidence
- Mathematics as a language and problem-solving tool
- Matter, energy and waves
- Forces and fields

In this Coursebook, the mathematics has been kept to the minimum required by the Cambridge International Examinations AS and A Level Physics syllabus. If you are also studying mathematics, you may find that more advanced techniques such as calculus will help you with many aspects of physics.

Studying physics can be a stimulating and worthwhile experience. It is an international subject; no single country has a monopoly on the development of the ideas. It can be a rewarding exercise to discover how men and women from many countries have contributed to our knowledge and well-being, through their research into and application of the concepts of physics. We hope not only that this book will help you to succeed in your future studies and career, but also that it will stimulate your curiosity and fire your imagination. Today’s students become the next generation of physicists and engineers, and we hope that you will learn from the past to take physics to ever-greater heights.

How to use this book

Each chapter begins with a short list of the facts and concepts that are explained in it.



There is a short context at the beginning of each chapter, containing an example of how the material covered in the chapter relates to the 'real world'.

Describing movement

Our eyes are good at detecting movement. We notice even quite small movements out of the corners of our eyes. It's important for us to be able to judge movement – think about crossing the road, cycling or driving, or catching a ball.

Figure 1.1 shows a way in which movement can be recorded on a photograph. This is a stroboscopic photograph of a boy juggling three balls. As he juggles, a bright lamp flashes several times a second so that the camera records the positions of the balls at equal intervals of time.

If we knew the time between flashes, we could measure the photograph and calculate the speed of a ball as it moves through the air.

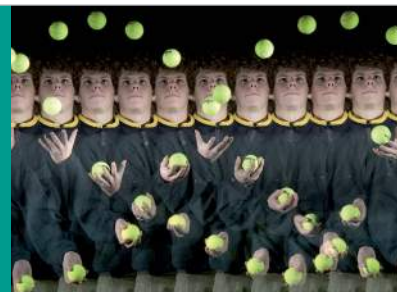


Figure 1.1 This boy is juggling three balls. A stroboscopic lamp flashes at regular intervals; the camera is moved to one side at a steady rate to show separate images of the boy.

The text and illustrations describe and explain all of the facts and concepts that you need to know. The chapters, and often the content within them as well, are arranged in a similar sequence to your syllabus, but with AS and A Level content clearly separated into the two halves of the book.

Figure 13.3 or a similar graph of displacement against time illustrates the following important definitions about waves and wave motion:

- The distance of a point on the wave from its undisturbed position or equilibrium position is called the **displacement** x .
- The maximum displacement of any point on the wave from its undisturbed position is called the **amplitude** A . The amplitude of a wave on the sea is measured in units of distance, e.g. metres. The greater the amplitude of the wave, the louder the sound or the rougher the sea!
- The distance from any point on a wave to the next exactly similar point (e.g. crest to crest) is called the **wavelength** λ (the Greek letter lambda). The wavelength of a wave on the sea is measured in units of distance, e.g. metres.
- The time taken for one complete oscillation of a point in a wave is called its **period** T . It is the time taken for a point to move from one particular position and return to that same position, moving in the same direction. It is measured in units of time, e.g. seconds.
- The number of oscillations per unit time of a point in a wave is called its **frequency** f . For sound waves, the higher the frequency of a musical note, the higher is its pitch. Frequency is measured in hertz (Hz), where 1 Hz = one oscillation per second (1 kHz = 10^3 Hz and 1 MHz = 10^6 Hz). The frequency f of a wave is the reciprocal of the period T :

$$f = \frac{1}{T}$$

Waves are called **mechanical waves** if they need a substance (medium) through which to travel. Sound is one example of such a wave. Other cases are waves on strings, seismic waves and water waves (Figure 13.4).

Some properties of typical waves are given on page 183 in Table 13.1.



Figure 13.4 The impact of a droplet on the surface of a liquid creates a vibration, which in turn gives rise to waves on the surface.

QUESTION

- Determine the wavelength and amplitude of each of the two waves shown in Figure 13.5.

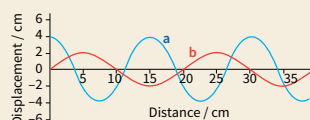


Figure 13.5 Two waves – for Question 1.

BOX 13.1: Measuring frequency

You can measure the frequency of sound waves using a cathode-ray oscilloscope (c.r.o.). Figure 13.6 shows how.

A microphone is connected to the input of the c.r.o. Sound waves are captured by the microphone and converted into a varying voltage which has the same frequency as the sound waves. This voltage is displayed on the c.r.o. screen.

It is best to think of a c.r.o. as a voltmeter which is capable of displaying a rapidly varying voltage. To do this, its spot moves across the screen at a steady speed, set by the time-base control. At the same time, the spot moves up and down according to the voltage of the input.

Hence the display on the screen is a graph of the varying voltage, with time on the (horizontal) x-axis. If we know the horizontal scale, we can determine the period and hence the frequency of the sound wave. Worked example 1 shows how to do this. (In Chapter 15 we will look at one method of measuring the wavelength of sound waves.)

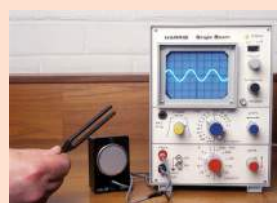


Figure 13.6 Measuring the frequency of sound waves from a tuning fork.

Questions throughout the text give you a chance to check that you have understood the topic you have just read about. You can find the answers to these questions on the CD-ROM.

This book does not contain detailed instructions for doing particular experiments, but you will find background information about the practical work you need to do in these Boxes. There are also two chapters, P1 and P2, which provide detailed information about the practical skills you need to develop during your course.

Important equations and other facts are shown in highlight boxes.

For an object of mass m travelling at a speed v , we have:

$$\text{kinetic energy} = \frac{1}{2} \times \text{mass} \times \text{speed}^2$$

$$E_k = \frac{1}{2}mv^2$$

Wherever you need to know how to use a formula to carry out a calculation, there are worked example boxes to show you how to do this.

WORKED EXAMPLE

- 1 In Figure 6.5, trolley A of mass 0.80 kg travelling at a velocity of 3.0 m s^{-1} collides head-on with a stationary trolley B. Trolley B has twice the mass of trolley A. The trolleys stick together and have a common velocity of 1.0 m s^{-1} after the collision. Show that momentum is conserved in this collision.

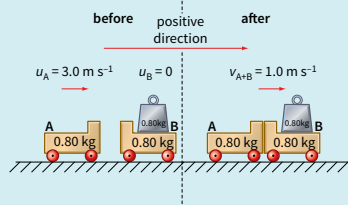


Figure 6.5 The state of trolleys A and B, before and after the collision.

Step 1 Make a sketch using the information given in the question. Notice that we need two diagrams to show the situations, one before and one after the collision.

Similarly, we need two calculations – one for the momentum of the trolleys before the collision and one for their momentum after the collision.

Step 2 Calculate the momentum before the collision:
momentum of trolleys before collision

$$\begin{aligned} &= m_A \times u_A + m_B \times u_B \\ &= (0.80 \times 3.0) + 0 \\ &= 2.4\text{ kg m s}^{-1} \end{aligned}$$

Trolley B has no momentum before the collision, because it is not moving.

Step 3 Calculate the momentum after the collision:
momentum of trolleys after collision

$$\begin{aligned} &= (m_A + m_B) \times v_{A+B} \\ &= (0.80 + 1.60) \times 1.0 \\ &= 2.4\text{ kg m s}^{-1} \end{aligned}$$

So, both before and after the collision, the trolleys have a combined momentum of 2.4 kg m s^{-1} . Momentum has been conserved.

Key words are highlighted in the text when they are first introduced.

The metre, kilogram and second are three of the seven SI **base units**. These are defined with great precision so that every standards laboratory can reproduce them correctly.

You will also find definitions of these words in the Glossary.

base units Defined units of the SI system from which all other units are derived.

There is a summary of key points at the end of each chapter. You might find this helpful when you are revising.

Summary

- Forces are vector quantities that can be added by means of a vector triangle. Their resultant can be determined using trigonometry or by scale drawing.
- Vectors such as forces can be resolved into components. Components at right angles to one another can be treated independently of one another. For a force F at an angle θ to the x -direction, the components are:
 x -direction: $F \cos \theta$
 y -direction: $F \sin \theta$
- The moment of a force = force $\times$ perpendicular distance of the pivot from the line of action of the force.
- The principle of moments states that, for any object that is in equilibrium, the sum of the clockwise moments about any point provided by the forces acting on the object equals the sum of the anticlockwise moments about that same point.
- A couple is a pair of equal, parallel but opposite forces whose effect is to produce a turning effect on a body without giving it linear acceleration.
torque of a couple = one of the forces $\times$ perpendicular distance between the forces
- For an object to be in equilibrium, the resultant force acting on the object must be zero and the resultant moment must be zero.

Questions at the end of each chapter begin with shorter answer questions, then move on to more demanding exam-style questions, some of which may require use of knowledge from previous chapters. Answers to these questions can be found on the CD-ROM.

- 1 Figure 15.19 shows a stationary wave on a string.

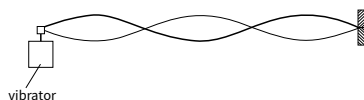


Figure 15.19 For End-of-chapter Question 1.

- a On a copy of Figure 15.19, label one **node** (N) and one **antinode** (A). [1]
 - b Mark on your diagram the wavelength of the standing wave and label it λ . [1]
 - c The frequency of the vibrator is doubled. Describe the changes in the standing wave pattern. [1]
- 2 A tuning fork which produces a note of 256 Hz is placed above a tube which is nearly filled with water. The water level is lowered until resonance is first heard.
- a Explain what is meant by the term **resonance**. [1]
 - b The length of the column of air above the water when resonance is first heard is 31.2 cm . Calculate the speed of the sound wave. [2]

Chapter 1: Kinematics – describing motion

Learning outcomes

You should be able to:

- define displacement, speed and velocity
- draw and interpret displacement–time graphs
- describe laboratory methods for determining speed
- use vector addition to add two or more vectors

Describing movement

Our eyes are good at detecting movement. We notice even quite small movements out of the corners of our eyes. It's important for us to be able to judge movement – think about crossing the road, cycling or driving, or catching a ball.

Figure 1.1 shows a way in which movement can be recorded on a photograph. This is a stroboscopic photograph of a boy juggling three balls. As he juggles, a bright lamp flashes several times a second so that the camera records the positions of the balls at equal intervals of time.

If we knew the time between flashes, we could measure the photograph and calculate the speed of a ball as it moves through the air.

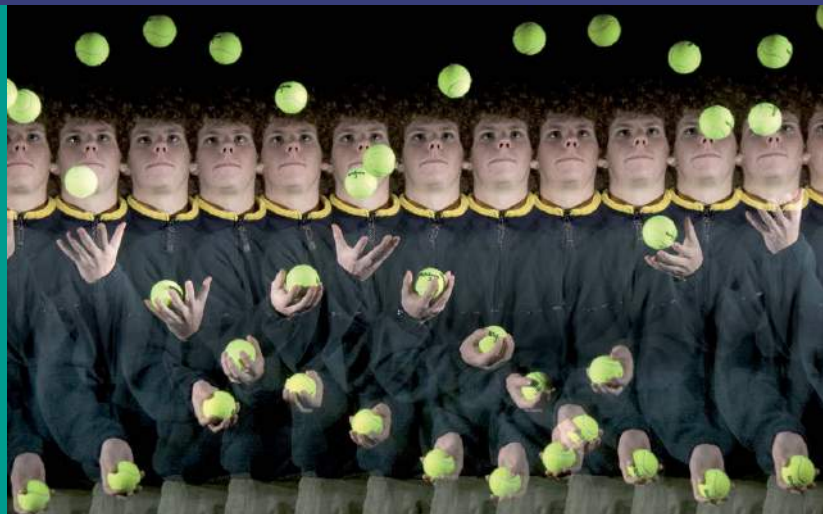


Figure 1.1 This boy is juggling three balls. A stroboscopic lamp flashes at regular intervals; the camera is moved to one side at a steady rate to show separate images of the boy.

Speed

We can calculate the average speed of something moving if we know the distance it moves and the time it takes:

$$\text{average speed} = \frac{\text{distance}}{\text{time}}$$

In symbols, this is written as:

$$v = \frac{d}{t}$$

where v is the average speed and d is the distance travelled in time t . The photograph (Figure 1.2) shows Ethiopia's Kenenisa Bekele posing next to the scoreboard after breaking the world record in a men's 10 000 metres race. The time on the clock in the photograph enables us to work out his average speed.

If the object is moving at a constant speed, this equation will give us its speed during the time taken. If its speed is changing, then the equation gives us its **average speed**. Average speed is calculated over a period of time.



Figure 1.2 Ethiopia's Kenenisa Bekele set a new world record for the 10 000 metres race in 2005.

If you look at the speedometer in a car, it doesn't tell you the car's average speed; rather, it tells you its speed at the instant when you look at it. This is the car's **instantaneous speed**.

QUESTION

- Look at Figure 1.2. The runner ran 10 000 m, and the clock shows the total time taken. Calculate his average speed during the race.

Units

In the *Système Internationale d'Unités* (the SI system), distance is measured in metres (m) and time in seconds (s). Therefore, speed is in metres per second. This is written as m s^{-1} (or as m/s). Here, s^{-1} is the same as 1/s, or 'per second'.

There are many other units used for speed. The choice of unit depends on the situation. You would probably give the speed of a snail in different units from the speed of a racing car. Table 1.1 includes some alternative units of speed.

Note that in many calculations it is necessary to work in SI units (m s^{-1}).

m s^{-1}	metres per second
cm s^{-1}	centimetres per second
km s^{-1}	kilometres per second
km h^{-1} or km/h	kilometres per hour
mph	miles per hour

Table 1.1 Units of speed.

QUESTIONS

- 2 Here are some units of speed:

ms^{-1} mms^{-1} kms^{-1} kmh^{-1}

Which of these units would be appropriate when stating the speed of each of the following?

- a a tortoise
- b a car on a long journey
- c light
- d a sprinter.

- 3 A snail crawls 12 cm in one minute. What is its average speed in mms^{-1} ?

Determining speed

You can find the speed of something moving by measuring the time it takes to travel between two fixed points. For example, some motorways have emergency telephones every 2000 m. Using a stopwatch you can time a car over this distance. Note that this can only tell you the car's average speed between the two points. You cannot tell whether it was increasing its speed, slowing down, or moving at a constant speed.

BOX 1.1: Laboratory measurements of speed

Here we describe four different ways to measure the speed of a trolley in the laboratory as it travels along a straight line. Each can be adapted to measure the speed of other moving objects, such as a glider on an air track, or a falling mass.

Measuring speed using two light gates

The leading edge of the card in Figure 1.3 breaks the light beam as it passes the first light gate. This starts the timer. The timer stops when the front of the card breaks the second beam. The trolley's speed is calculated from the time interval and the distance between the light gates.

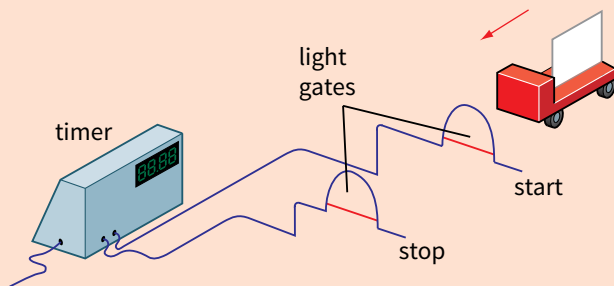


Figure 1.3 Using two light gates to find the average speed of a trolley.

Measuring speed using one light gate

The timer in Figure 1.4 starts when the leading edge of the card breaks the light beam. It stops when the trailing edge passes through. In this case, the time shown is the time taken for the trolley to travel a distance equal to the length of the card. The computer software can calculate the speed directly by dividing the distance by the time taken.

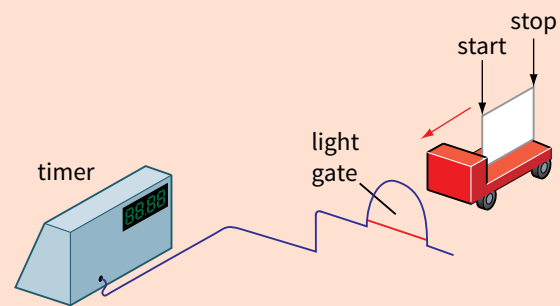


Figure 1.4 Using a single light gate to find the average speed of a trolley.

Measuring speed using a ticker-timer

The ticker-timer (Figure 1.5) marks dots on the tape at regular intervals, usually s (i.e. 0.02 s). (This is because it works with alternating current, and in most countries the frequency of the alternating mains is 50 Hz.) The pattern of dots acts as a record of the trolley's movement.

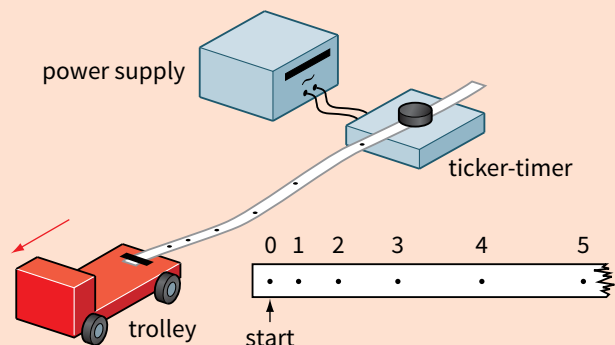


Figure 1.5 Using a ticker-timer to investigate the motion of a trolley.

BOX 1.1: Laboratory measurements of speed (continued)

Start by inspecting the tape. This will give you a description of the trolley's movement. Identify the start of the tape. Then look at the spacing of the dots:

- even spacing – constant speed
- increasing spacing – increasing speed.

Now you can make some measurements. Measure the distance of every fifth dot from the start of the tape. This will give you the trolley's distance at intervals of 0.1 s. Put the measurements in a table and draw a distance–time graph.

Measuring speed using a motion sensor

The motion sensor (Figure 1.6) transmits regular pulses of ultrasound at the trolley. It detects the reflected waves and determines the time they took for the trip to the trolley and back. From this, the computer can deduce the distance to the trolley from the motion sensor. It can generate a distance–time graph. You can determine the speed of the trolley from this graph.

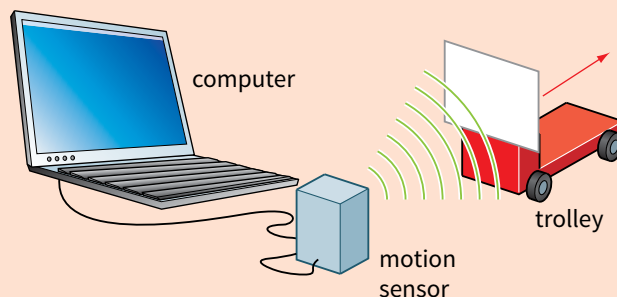


Figure 1.6 Using a motion sensor to investigate the motion of a trolley.

Choosing the best method

Each of these methods for finding the speed of a trolley has its merits. In choosing a method, you might think about the following points:

- Does the method give an average value of speed or can it be used to give the speed of the trolley at different points along its journey?
- How precisely does the method measure time – to the nearest millisecond?
- How simple and convenient is the method to set up in the laboratory?

QUESTIONS

- 4 A trolley with a 5.0 cm long card passed through a single light gate. The time recorded by a digital timer was 0.40 s. What was the average speed of the trolley in ms^{-1} ?
- 5 Figure 1.7 shows two ticker-tapes. Describe the motion of the trolleys which produced them.

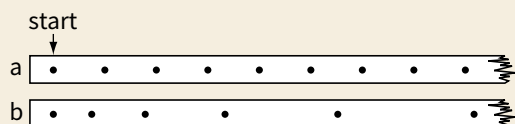


Figure 1.7 Two ticker-tapes; for Question 5.

- 6 Four methods for determining the speed of a moving trolley have been described. Each could be adapted to investigate the motion of a falling mass. Choose two methods which you think would be suitable, and write a paragraph for each to say how you would adapt it for this purpose.

Distance and displacement, scalar and vector

In physics, we are often concerned with the distance moved by an object in a particular direction. This is called its **displacement**. Figure 1.8 illustrates the difference between distance and displacement. It shows the route followed by walkers as they went from town A to town C. Their winding route took them through town B, so that they covered a total distance of 15 km. However, their displacement was much less than this. Their finishing position was just 10 km from where they started. To give a complete statement of their displacement, we need to give both distance and direction:

$$\text{displacement} = 10 \text{ km } 30^\circ \text{ E of N}$$

Displacement is an example of a **vector quantity**. A vector quantity has both magnitude (size) and direction. Distance, on the other hand, is a **scalar quantity**. Scalar quantities have magnitude only.

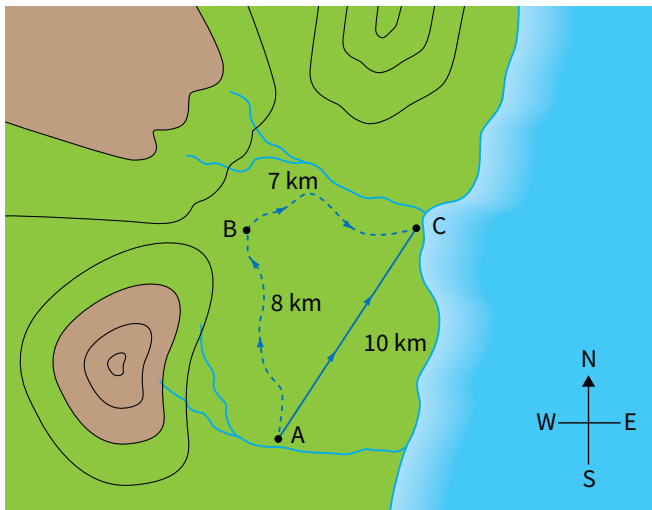


Figure 1.8 If you go on a long walk, the distance you travel will be greater than your displacement. In this example, the walkers travel a distance of 15 km, but their displacement is only 10 km, because this is the distance from the start to the finish of their walk.

Speed and velocity

It is often important to know both the speed of an object and the direction in which it is moving. Speed and direction are combined in another quantity, called **velocity**. The velocity of an object can be thought of as its speed in a particular direction. So, like displacement, velocity is a **vector** quantity. Speed is the corresponding scalar quantity, because it does not have a direction. So, to give the velocity of something, we have to state the direction in which it is moving. For example, an aircraft flies with a velocity of 300 m s^{-1} due north. Since velocity is a vector quantity, it is defined in terms of displacement:

$$\text{velocity} = \frac{\text{change in displacement}}{\text{time taken}}$$

Alternatively, we can say that velocity is the rate of change of an object's displacement. From now on, you need to be clear about the distinction between velocity and speed, and between displacement and distance. **Table 1.2** shows the standard symbols and units for these quantities.

Quantity	Symbol for quantity	Symbol for unit
distance	d	m
displacement	s, x	m
time	t	s
speed, velocity	v	m s^{-1}

Table 1.2 Standard symbols and units. (Take care not to confuse italic s for displacement with s for seconds. Notice also that v is used for both speed and velocity.)

QUESTION

- 7 Which of these gives speed, velocity, distance or displacement? (Look back at the definitions of these quantities.)
- a The ship sailed south-west for 200 miles.
 - b I averaged 7 mph during the marathon.
 - c The snail crawled at 2 mm s^{-1} along the straight edge of a bench.
 - d The sales representative's round trip was 420 km.

Speed and velocity calculations

We can write the equation for velocity in symbols:

$$v = \frac{s}{t}$$

$$v = \frac{\Delta s}{\Delta t}$$

The word equation for velocity is:

$$\text{velocity} = \frac{\text{change in displacement}}{\text{time taken}}$$

Note that we are using Δs to mean 'change in displacement s '. The symbol Δ , Greek letter delta, means 'change in'. It does not represent a quantity (in the way that s does); it is simply a convenient way of representing a change in a quantity. Another way to write Δs would be $s_2 - s_1$, but this is more time-consuming and less clear.

The equation for velocity, $v = \frac{\Delta s}{\Delta t}$, can be rearranged as follows, depending on which quantity we want to determine:

$$\text{change in displacement } \Delta s = v \times \Delta t$$

$$\text{change in time } \Delta t = \frac{\Delta s}{v}$$

Note that each of these equations is balanced in terms of units. For example, consider the equation for displacement. The units on the right-hand side are $\text{m s}^{-1} \times \text{s}$, which simplifies to m, the correct unit for displacement.

Note also that we can, of course, use the same equations to find speed and distance, that is:

$$v = \frac{d}{t}$$

$$\text{distance } d = v \times t$$

$$\text{time } t = \frac{d}{v}$$

WORKED EXAMPLES

- 1** A car is travelling at 15 m s^{-1} . How far will it travel in 1 hour?

Step 1 It is helpful to start by writing down what you know and what you want to know:

$$v = 15 \text{ m s}^{-1}$$

$$t = 1 \text{ h} = 3600 \text{ s}$$

$$d = ?$$

Step 2 Choose the appropriate version of the equation and substitute in the values. Remember to include the units:

$$d = v \times t$$

$$= 15 \times 3600$$

$$= 5.4 \times 10^4 \text{ m}$$

$$= 54 \text{ km}$$

The car will travel 54 km in 1 hour.

- 2** The Earth orbits the Sun at a distance of 150 000 000 km. How long does it take light from the Sun to reach the Earth? (Speed of light in space = $3.0 \times 10^8 \text{ m s}^{-1}$.)

Step 1 Start by writing what you know. Take care with units; it is best to work in m and s. You need to be able to express numbers in scientific notation (using powers of 10) and to work with these on your calculator.

$$v = 3.0 \times 10^8 \text{ m s}^{-1}$$

$$d = 150\,000\,000 \text{ km}$$

$$= 150\,000\,000\,000 \text{ m}$$

$$= 1.5 \times 10^{11} \text{ m}$$

Step 2 Substitute the values in the equation for time:

$$t = \frac{d}{v} = \frac{1.5 \times 10^{11}}{3.0 \times 10^8} = 500 \text{ s}$$

Light takes 500 s (about 8.3 minutes) to travel from the Sun to the Earth.

Hint: When using a calculator, to calculate the time t , you press the buttons in the following sequence:

[1.5] [EXP] [11] [$\div$] [3] [EXP] [8]

or

[1.5] [$\times 10^n$] [11] [$\div$] [3] [$\times 10^n$] [8]

Making the most of units

In Worked example 1 and Worked example 2, units have been omitted in intermediate steps in the calculations. However, at times it can be helpful to include units as this can be a way of checking that you have used the correct equation; for example, that you have not divided one quantity by another when you should have multiplied them. The units of an equation must be balanced, just as the numerical values on each side of the equation must be equal.

If you take care with units, you should be able to carry out calculations in non-SI units, such as kilometres per hour, without having to convert to metres and seconds.

For example, how far does a spacecraft travelling at $40\,000 \text{ km h}^{-1}$ travel in one day? Since there are 24 hours in one day, we have:

$$\begin{aligned} \text{distance travelled} &= 40\,000 \text{ km h}^{-1} \times 24 \text{ h} \\ &= 960\,000 \text{ km} \end{aligned}$$

QUESTIONS

- 8** A submarine uses sonar to measure the depth of water below it. Reflected sound waves are detected 0.40 s after they are transmitted. How deep is the water? (Speed of sound in water = 1500 m s^{-1} .)
- 9** The Earth takes one year to orbit the Sun at a distance of $1.5 \times 10^{11} \text{ m}$. Calculate its speed. Explain why this is its average speed and not its velocity.

Displacement–time graphs

We can represent the changing position of a moving object by drawing a displacement–time graph. The gradient (slope) of the graph is equal to its velocity (Figure 1.9). The steeper the slope, the greater the velocity. A graph like this can also tell us if an object is moving forwards or backwards. If the gradient is negative, the object's velocity is negative – it is moving backwards.

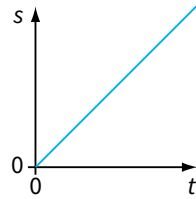
Deducing velocity from a displacement–time graph

A toy car moves along a straight track. Its displacement at different times is shown in Table 1.3. This data can be used to draw a displacement–time graph from which we can deduce the car's velocity.

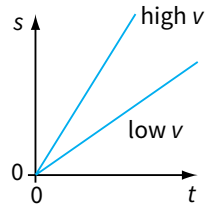
Displacement / m	1.0	3.0	5.0	7.0	7.0	7.0
Time / s	0.0	1.0	2.0	3.0	4.0	5.0

Table 1.3 Displacement (s) and time (t) data for a toy car.

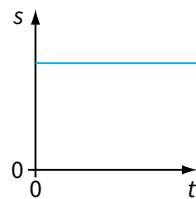
The straight line shows that the object's velocity is constant.



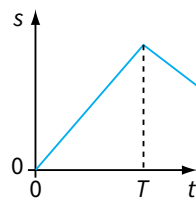
The slope shows which object is moving faster. The steeper the slope, the greater the velocity.



The slope of this graph is 0. The displacement s is not changing. Hence the velocity $v = 0$. The object is stationary.



The slope of this graph suddenly becomes negative. The object is moving back the way it came. Its velocity v is negative after time T .



This displacement-time graph is curved. The slope is changing. This means that the object's velocity is changing – this is considered in Chapter 2.

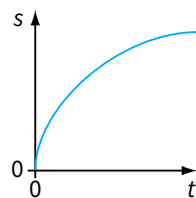


Figure 1.9 The slope of a displacement–time (s – t) graph tells us how fast an object is moving.

It is useful to look at the data first, to see the pattern of the car's movement. In this case, the displacement increases steadily at first, but after 3.0 s it becomes constant. In other words, initially the car is moving at a steady velocity, but then it stops.

Now we can plot the displacement–time graph (Figure 1.11).

We want to work out the velocity of the car over the first 3.0 seconds. We can do this by working out the gradient of the graph, because:

velocity = gradient of displacement–time graph

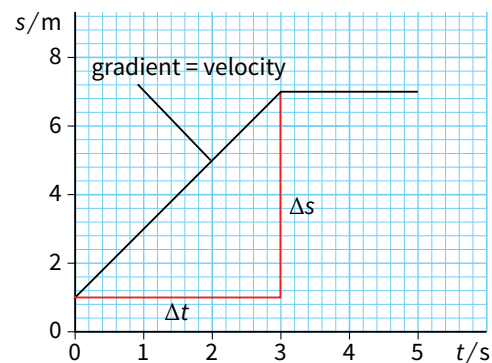


Figure 1.11 Displacement–time graph for a toy car; data as shown in Table 1.3.

We draw a right-angled triangle as shown. To find the car's velocity, we divide the change in displacement by the change in time. These are given by the two sides of the triangle labelled Δs and Δt .

$$\begin{aligned} \text{velocity } v &= \frac{\text{change in displacement}}{\text{time taken}} \\ v &= \frac{\Delta s}{\Delta t} \\ v &= \frac{(7.0 - 1.0)}{(3.0 - 0)} = \frac{6.0}{3.0} = 2.0 \text{ ms}^{-1} \end{aligned}$$

If you are used to finding the gradient of a graph, you may be able to reduce the number of steps in this calculation.

QUESTIONS

- 10 The displacement–time sketch graph in Figure 1.10 represents the journey of a bus. What does the graph tell you about the journey?

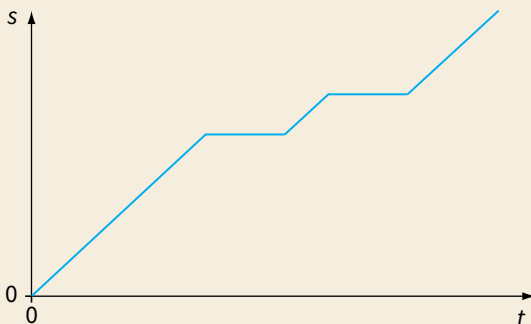


Figure 1.10 For Question 10.

- 11 Sketch a displacement–time graph to show your motion for the following event. You are walking at a constant speed across a field after jumping off a gate. Suddenly you see a bull and stop. Your friend says there's no danger, so you walk on at a reduced constant speed. The bull bellows, and you run back to the gate. Explain how each section of the walk relates to a section of your graph.
- 12 Table 1.4 shows the displacement of a racing car at different times as it travels along a straight track during a speed trial.
- Determine the car's velocity.
 - Draw a displacement–time graph and use it to find the car's velocity.

Displacement / m	0	85	170	255	340
Time / s	0	1.0	2.0	3.0	4.0

Table 1.4 Displacement (s) and time (t) data for Question 12.

- 13 An old car travels due south. The distance it travels at hourly intervals is shown in Table 1.5.
- Draw a distance–time graph to represent the car's journey.
 - From the graph, deduce the car's speed in km h^{-1} during the first three hours of the journey.
 - What is the car's average speed in km h^{-1} during the whole journey?

Time / h	Distance / km
0	0
1	23
2	46
3	69
4	84

Table 1.5 Data for Question 13.

Combining displacements

The walkers shown in Figure 1.12 are crossing difficult ground. They navigate from one prominent point to the next, travelling in a series of straight lines. From the map, they can work out the distance that they travel and their displacement from their starting point:

distance travelled = 25 km

(Lay thread along route on map; measure thread against map scale.)

displacement = 15 km north-east

(Join starting and finishing points with straight line; measure line against scale.)

A map is a scale drawing. You can find your displacement by measuring the map. But how can you **calculate** your displacement? You need to use ideas from geometry and trigonometry. Worked examples 3 and 4 show how.

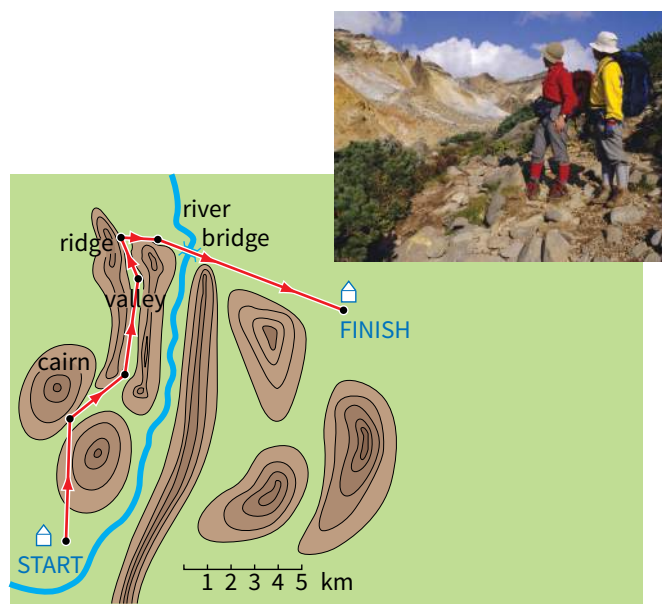


Figure 1.12 In rough terrain, walkers head straight for a prominent landmark.

WORKED EXAMPLES

- 3 A spider runs along two sides of a table (Figure 1.13). Calculate its final displacement.

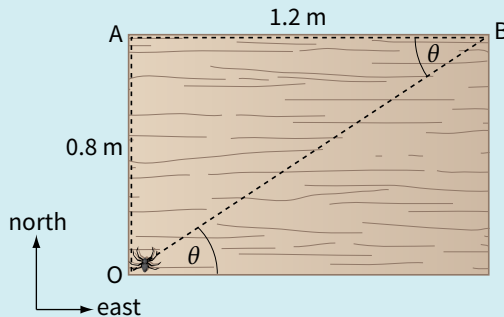


Figure 1.13 The spider runs a distance of 2.0 m, but what is its displacement?

Step 1 Because the two sections of the spider's run (OA and AB) are at right angles, we can **add** the two displacements using Pythagoras's theorem:

$$\begin{aligned} OB^2 &= OA^2 + AB^2 \\ &= 0.8^2 + 1.2^2 = 2.08 \\ OB &= \sqrt{2.08} = 1.44 \text{ m} \approx 1.4 \text{ m} \end{aligned}$$

Step 2 Displacement is a vector. We have found the **magnitude** of this vector, but now we have to find its direction. The angle θ is given by:

$$\begin{aligned} \tan \theta &= \frac{\text{opp}}{\text{adj}} = \frac{0.8}{1.2} \\ &= 0.667 \\ \theta &= \tan^{-1}(0.667) \\ &= 33.7^\circ \approx 34^\circ \end{aligned}$$

So the spider's displacement is 1.4 m at an angle of 34° north of east.

- 4 An aircraft flies 30 km due east and then 50 km north-east (Figure 1.14). Calculate the final displacement of the aircraft.

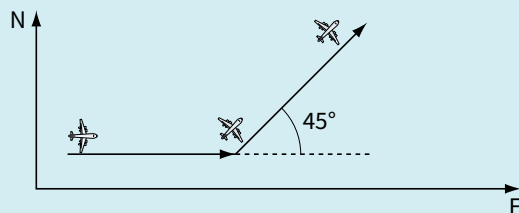


Figure 1.14 What is the aircraft's final displacement?

Here, the two displacements are not at 90° to one another, so we can't use Pythagoras's theorem. We can solve this problem by making a scale drawing, and measuring the final displacement. (However, you could solve the same problem using trigonometry.)

Step 1 Choose a suitable scale. Your diagram should be reasonably large; in this case, a scale of 1 cm to represent 5 km is reasonable.

Step 2 Draw a line to represent the first vector. North is at the top of the page. The line is 6 cm long, towards the east (right).

Step 3 Draw a line to represent the second vector, starting at the end of the first vector. The line is 10 cm long, and at an angle of 45° (Figure 1.15).

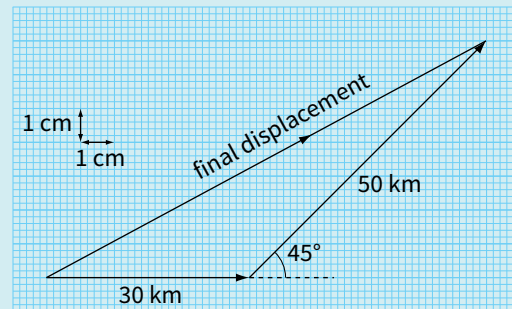


Figure 1.15 Scale drawing for Worked example 4. Using graph paper can help you to show the vectors in the correct directions.

Step 4 To find the final displacement, join the start to the finish. You have created a **vector triangle**. Measure this displacement vector, and use the scale to convert back to kilometres:

length of vector = 14.8 cm

final displacement = $14.8 \times 5 = 74 \text{ km}$

Step 5 Measure the angle of the final displacement vector:

angle = 28° N of E

Therefore the aircraft's final displacement is 74 km at 28° north of east.

QUESTIONS

- 14** You walk 3.0 km due north, and then 4.0 km due east.
- Calculate the total distance in km you have travelled.
 - Make a scale drawing of your walk, and use it to find your final displacement. Remember to give both the magnitude and the direction.
 - Check your answer to part **b** by calculating your displacement.
- 15** A student walks 8.0 km south-east and then 12 km due west.
- Draw a vector diagram showing the route. Use your diagram to find the total displacement. Remember to give the scale on your diagram and to give the direction as well as the magnitude of your answer.
 - Calculate the resultant displacement. Show your working clearly.

This process of adding two displacements together (or two or more of any type of vector) is known as **vector addition**. When two or more vectors are added together, their combined effect is known as the **resultant** of the vectors.

Combining velocities

Velocity is a vector quantity and so two velocities can be combined by vector addition in the same way that we have seen for two or more displacements.

Imagine that you are attempting to swim across a river. You want to swim directly across to the opposite bank, but the current moves you sideways at the same time as you are swimming forwards. The outcome is that you will end up on the opposite bank, but downstream of your intended landing point. In effect, you have two velocities:

- the velocity due to your swimming, which is directed straight across the river
- the velocity due to the current, which is directed downstream, at right angles to your swimming velocity.

These combine to give a **resultant** (or net) velocity, which will be diagonally downstream. In order to swim directly across the river, you would have to aim upstream. Then your resultant velocity could be directly across the river.

WORKED EXAMPLE

- 5** An aircraft is flying due north with a velocity of 200 m s^{-1} . A side wind of velocity 50 m s^{-1} is blowing due east. What is the aircraft's resultant velocity (give the magnitude and direction)?

Here, the two velocities are at 90° . A sketch diagram and Pythagoras's theorem are enough to solve the problem.

Step 1 Draw a sketch of the situation – this is shown in Figure 1.16a.

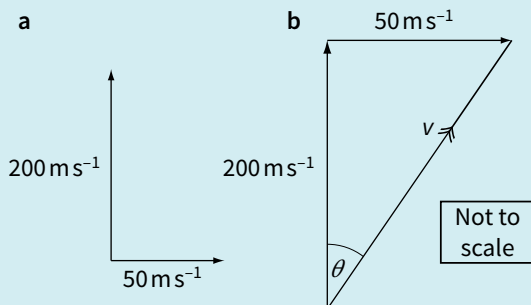


Figure 1.16 Finding the resultant of two velocities – for Worked example 5.

Step 2 Now sketch a vector triangle. Remember that the second vector starts where the first one ends. This is shown in Figure 1.16b.

Step 3 Join the start and end points to complete the triangle.

Step 4 Calculate the magnitude of the resultant vector v (the hypotenuse of the right-angled triangle).

$$v^2 = 200^2 + 50^2 = 40\,000 + 2\,500 = 42\,500$$

$$v = \sqrt{42\,500} \approx 206 \text{ m s}^{-1}$$

Step 5 Calculate the angle θ :

$$\tan \theta = \frac{50}{200}$$

$$= 0.25$$

$$\theta = \tan^{-1}(0.25) \approx 14^\circ$$

So the aircraft's resultant velocity is 206 m s^{-1} at 14° east of north.

QUESTIONS

- 16** A swimmer can swim at 2.0 m s^{-1} in still water. She aims to swim directly across a river which is flowing at 0.80 m s^{-1} . Calculate her resultant velocity. (You must give both the magnitude and the direction.)
- 17** A stone is thrown from a cliff and strikes the surface of the sea with a vertical velocity of 18 m s^{-1} and a horizontal velocity v . The resultant of these two velocities is 25 m s^{-1} .
- Draw a vector diagram showing the two velocities and the resultant.
 - Use your diagram to find the value of v .
 - Use your diagram to find the angle between the stone and the vertical as it strikes the water.

Summary

- Displacement is the distance travelled in a particular direction.
- Velocity is defined by the word equation

$$\text{velocity} = \frac{\text{change in displacement}}{\text{time taken}}$$

The gradient of a displacement–time graph is equal to velocity:

$$\text{velocity} = \frac{\Delta s}{\Delta t}$$

- Distance and speed are scalar quantities. A scalar quantity has only magnitude.
- Displacement and velocity are vector quantities. A vector quantity has both magnitude and direction.
- Vector quantities may be combined by vector addition to find their resultant.

End-of-chapter questions

- 1 A car travels one complete lap around a circular track at a constant speed of 120 km h^{-1} .

- If one lap takes 2.0 minutes, show that the length of the track is 4.0 km. [2]
- Explain why values for the **average speed** and **average velocity** are different. [1]
- Determine the magnitude of the displacement of the car in a time of 1.0 minute. [2]
(The circumference of a circle = $2\pi R$, where R is the radius of the circle.)

- 2 A boat leaves point A and travels in a straight line to point B (Figure 1.17). The journey takes 60 s.

Calculate:

- the distance travelled by the boat [2]
- the total displacement of the boat [2]
- the average velocity of the boat. [2]

Remember that each vector quantity must be given a direction as well as a magnitude.

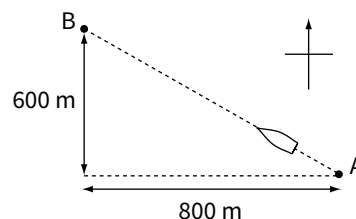


Figure 1.17 For End-of-chapter Question 2.

- 3 A boat travels at 2.0 m s^{-1} east towards a port, 2.2 km away. When the boat reaches the port, the passengers travel in a car due north for 15 minutes at 60 km h^{-1} .

Calculate:

- the total distance travelled [2]
 - the total displacement [2]
 - the total time taken [2]
 - the average speed in m s^{-1} [2]
 - the magnitude of the average velocity. [2]
- 4 A river flows from west to east with a constant velocity of 1.0 m s^{-1} . A boat leaves the south bank heading due north at 2.40 m s^{-1} . Find the resultant velocity of the boat. [2]
- 5
 - Define **displacement**. [1]
 - Use the definition of displacement to explain how it is possible for an athlete to run round a track yet have no displacement. [2]
- 6 A girl is riding a bicycle at a constant velocity of 3.0 m s^{-1} along a straight road. At time $t = 0$, she passes a boy sitting on a stationary bicycle. At time $t = 0$, the boy sets off to catch up with the girl. His velocity increases from time $t = 0$ until $t = 5.0 \text{ s}$, when he has covered a distance of 10 m. He then continues at a constant velocity of 4.0 m s^{-1} .
- Draw the displacement–time graph for the girl from $t = 0$ to 12 s. [1]
 - On the same graph axes, draw the displacement–time graph for the boy. [2]
 - Using your graph, determine the value of t when the boy catches up with the girl. [1]

- 7 A student drops a small black sphere alongside a vertical scale marked in centimetres. A number of flash photographs of the sphere are taken at 0.1 s intervals, as shown in Figure 1.18. The first photograph is taken with the sphere at the top at time $t = 0$ s.

- Explain how Figure 1.18 shows that the sphere reaches a constant speed.
- Determine the constant speed reached by the sphere.
- Determine the distance that the sphere has fallen when $t = 0.8$ s.

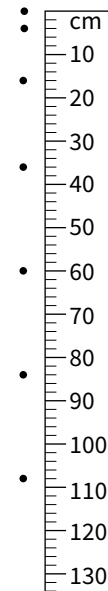


Figure 1.18 For End-of-chapter Question 7.

[2]
[2]
[2]

- State **one** difference between a scalar quantity and a vector quantity and give an example of each.
 - A plane has an air speed of 500 km h^{-1} due north. A wind blows at 100 km h^{-1} from east to west. Draw a vector diagram to calculate the resultant velocity of the plane. Give the direction of travel of the plane with respect to north.
 - The plane flies for 15 minutes. Calculate the displacement of the plane in this time.
- A small aircraft for one person is used on a short horizontal flight. On its journey from A to B, the resultant velocity of the aircraft is 15 m s^{-1} in a direction 60° east of north and the wind velocity is 7.5 m s^{-1} due north (Figure 1.19).

[3]
[4]
[1]

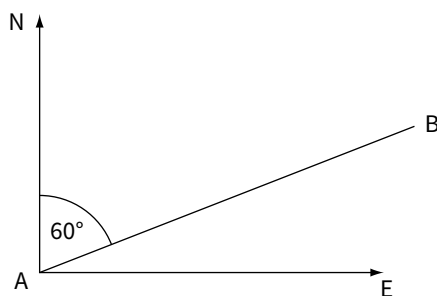


Figure 1.19 For End-of-chapter Question 9.

- Show that for the aircraft to travel from A to B it should be pointed due east.
- After flying 5 km from A to B, the aircraft returns along the same path from B to A with a resultant velocity of 13.5 m s^{-1} . Assuming that the time spent at B is negligible, calculate the average speed for the complete journey from A to B and back to A.

[2]
[3]

Chapter 2: Accelerated motion

Learning outcomes

You should be able to:

- define acceleration
- draw and interpret velocity–time graphs
- derive and use the equations of uniformly accelerated motion
- describe a method for determining the acceleration due to gravity, g
- explain projectile motion in terms of horizontal and vertical components of motion

Quick off the mark

The cheetah (Figure 2.1) has a maximum speed of over 30 m s^{-1} (108 km/h). From a standing start a cheetah can reach 20 m s^{-1} in just three or four strides, taking only two seconds.

A car cannot increase its speed as rapidly but on a long straight road it can easily travel faster than a cheetah.



Figure 2.1 The cheetah is the world's fastest land animal. Its acceleration is impressive, too.

The meaning of acceleration

In everyday language, the term **accelerating** means 'speeding up'. Anything whose speed is increasing is accelerating. Anything whose speed is decreasing is decelerating.

To be more precise in our definition of acceleration, we should think of it as **changing velocity**. Any object whose speed is changing or which is changing its **direction** has **acceleration**. Because acceleration is linked to velocity in this way, it follows that it is a **vector** quantity.

Some examples of objects accelerating are shown in Figure 2.2.

Calculating acceleration

The acceleration of something indicates the rate at which its velocity is changing. Language can get awkward here. Looking at the sprinter in Figure 2.3, we might say, 'The sprinter accelerates **faster** than the car.' However, 'faster' really means 'greater speed'. It is better to say, 'The sprinter has a greater acceleration than the car.'

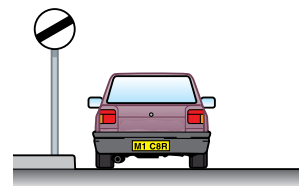
Acceleration is defined as follows:

$$\begin{aligned} \text{acceleration} &= \text{rate of change of velocity} \\ \text{average acceleration} &= \frac{\text{change in velocity}}{\text{time taken}} \end{aligned}$$

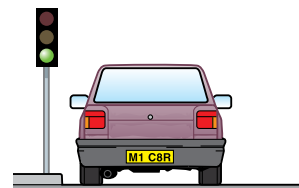
So to calculate acceleration a , we need to know two quantities – the change in velocity Δv and the time taken Δt :

$$a = \frac{\Delta v}{\Delta t}$$

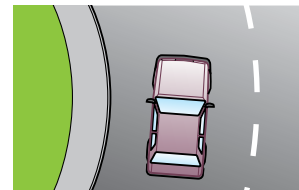
Sometimes this equation is written differently. We write u for the **initial velocity** and v for the **final velocity** (because u comes before v in the alphabet). The moving object



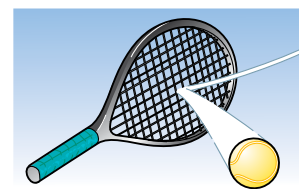
A car speeding up as it leaves the town. The driver presses on the accelerator pedal to increase the car's velocity.



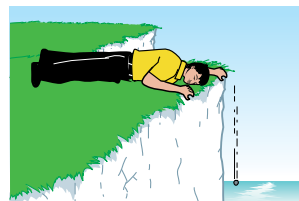
A car setting off from the traffic lights. There is an instant when the car is both stationary **and** accelerating. Otherwise it would not start moving.



A car travelling round a bend at a steady speed. The car's speed is constant, but its velocity is changing as it changes direction.



A ball being hit by a tennis racket. Both the ball's speed and direction are changing. The ball's velocity changes.



A stone dropped over a cliff. Gravity makes the stone go faster and faster. The stone accelerates as it falls.

Figure 2.2 Examples of objects accelerating.

accelerates from u to v in a time t (this is the same as the time represented by Δt above). Then the acceleration is given by the equation:

$$a = \frac{v - u}{t}$$

You must learn the definition of acceleration. It can be put in words or symbols. If you use symbols you must state what those symbols mean.

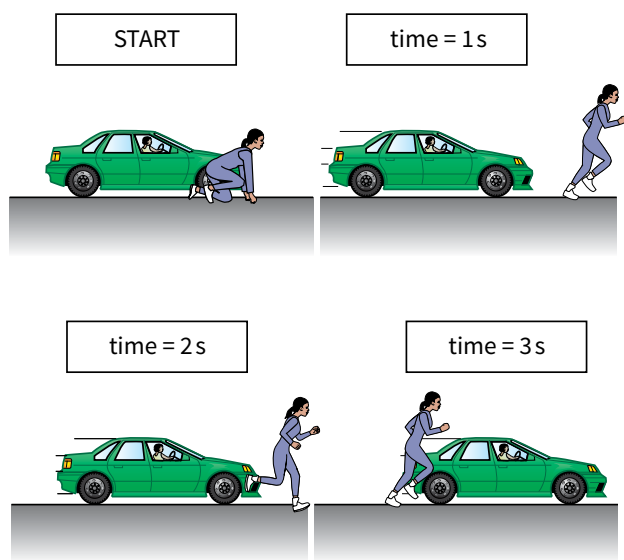


Figure 2.3 The sprinter has a greater acceleration than the car, but her top speed is less.

Units of acceleration

The unit of acceleration is m s^{-2} (metres per second squared). The sprinter might have an acceleration of 5 m s^{-2} ; her velocity increases by 5 m s^{-1} every second. You could express acceleration in other units. For example, an advertisement might claim that a car accelerates from 0 to 60 miles per hour (mph) in 10 s. Its acceleration would then be 6 mph s^{-1} (6 miles per hour per second). However, mixing together hours and seconds is not a good idea, and so acceleration is almost always given in the standard SI unit of m s^{-2} .

QUESTIONS

- 1 A car accelerates from a standing start and reaches a velocity of 18 m s^{-1} after 6.0 s. Calculate its acceleration.
- 2 A car driver brakes gently. Her car slows down from 23 m s^{-1} to 11 m s^{-1} in 20 s. Calculate the magnitude (size) of her deceleration. (Note that, because she is slowing down, her acceleration is negative.)
- 3 A stone is dropped from the top of a cliff. Its acceleration is 9.81 m s^{-2} . How fast is it moving:
 - a after 1 s?
 - b after 3 s?

WORKED EXAMPLES

- 1 Leaving a bus stop, a bus reaches a velocity of 8.0 m s^{-1} after 10 s. Calculate the acceleration of the bus.

Step 1 Note that the bus's initial velocity is 0 m s^{-1} . Therefore:

change in velocity $\Delta v = (8.0 - 0) \text{ m s}^{-1}$

time taken $\Delta t = 10 \text{ s}$

Step 2 Substitute these values in the equation for acceleration:

$$\begin{aligned} \text{acceleration} &= \frac{\Delta v}{\Delta t} = \frac{8.0}{10} \\ &= 0.80 \text{ m s}^{-2} \end{aligned}$$

- 2 A sprinter starting from rest has an acceleration of 5.0 m s^{-2} during the first 2.0 s of a race. Calculate her velocity after 2.0 s.

Step 1 Rearranging the equation $a = \frac{v - u}{t}$ gives:

$$v = u + at$$

Step 2 Substituting the values and calculating gives:

$$v = 0 + (5.0 \times 2.0) = 10 \text{ m s}^{-1}$$

- 3 A train slows down from 60 m s^{-1} to 20 m s^{-1} in 50 s. Calculate the magnitude of the deceleration of the train.

Step 1 Write what you know:

$$u = 60 \text{ m s}^{-1} \quad v = 20 \text{ m s}^{-1} \quad t = 50 \text{ s}$$

Step 2 Take care! Here the train's final velocity is less than its initial velocity. To ensure that we arrive at the correct answer, we will use the alternative form of the equation to calculate a .

$$\begin{aligned} a &= \frac{v - u}{t} \\ &= \frac{20 - 60}{50} = \frac{-40}{50} = -0.80 \text{ m s}^{-2} \end{aligned}$$

The minus sign (negative acceleration) indicates that the train is slowing down. It is decelerating. The magnitude of the deceleration is 0.80 m s^{-2} .

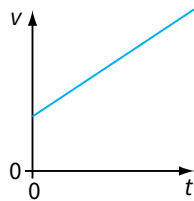
Deducing acceleration

The gradient of a velocity–time graph tells us whether the object’s velocity has been changing at a high rate or a low rate, or not at all (Figure 2.4). We can deduce the value of the acceleration from the gradient of the graph:

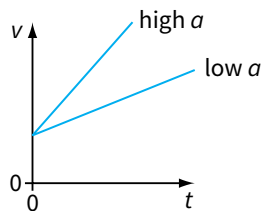
acceleration = gradient of velocity–time graph

The graph (Figure 2.5) shows how the velocity of a cyclist changed during the start of a sprint race. We can find his acceleration during the first section of the graph (where the line is straight) using the triangle as shown.

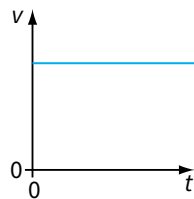
A straight line with a positive slope shows constant acceleration.



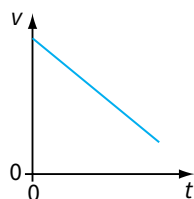
The greater the slope, the greater the acceleration.



The velocity is constant. Therefore acceleration $a = 0$.



A negative slope shows deceleration (a is negative).



The slope is changing; the acceleration is changing.

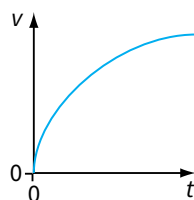


Figure 2.4 The gradient of a velocity–time graph is equal to acceleration.

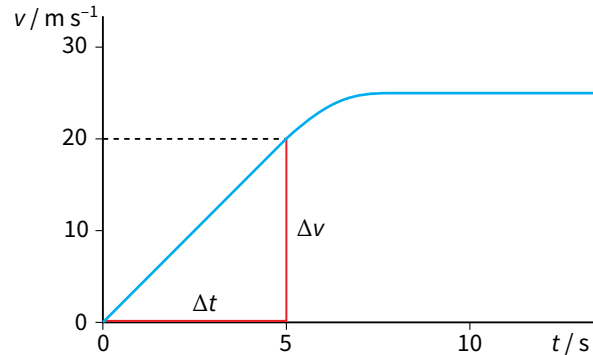


Figure 2.5 Deducing acceleration from a velocity–time graph.

The change in velocity Δv is given by the vertical side of the triangle. The time taken Δt is given by the horizontal side.

$$\begin{aligned}\text{acceleration} &= \frac{\text{change in displacement}}{\text{time taken}} \\ &= \frac{20 - 0}{5} \\ &= 4.0 \text{ m s}^{-2}\end{aligned}$$

A more complex example where the velocity–time graph is curved is shown on page 24.

Deducing displacement

We can also find the displacement of a moving object from its velocity–time graph. This is given by the area under the graph:

displacement = area under velocity–time graph

It is easy to see why this is the case for an object moving at a constant velocity. The displacement is simply velocity $\times$ time, which is the area of the shaded rectangle (Figure 2.6a).

For changing velocity, again the area under the graph gives displacement (Figure 2.6b). The area of each square of the graph represents a distance travelled: in this case, $1 \text{ m s}^{-1} \times 1 \text{ s}$, or 1 m . So, for this simple case in which the area is a triangle, we have:

$$\begin{aligned}\text{displacement} &= \frac{1}{2} \times \text{base} \times \text{height} \\ &= \frac{1}{2} \times 5.0 \times 10 = 25 \text{ m}\end{aligned}$$

It is easy to confuse displacement–time graphs and velocity–time graphs. Check by looking at the quantity marked on the vertical axis.

For more complex graphs, you may have to use other techniques such as counting squares to deduce the area, but this is still equal to the displacement.

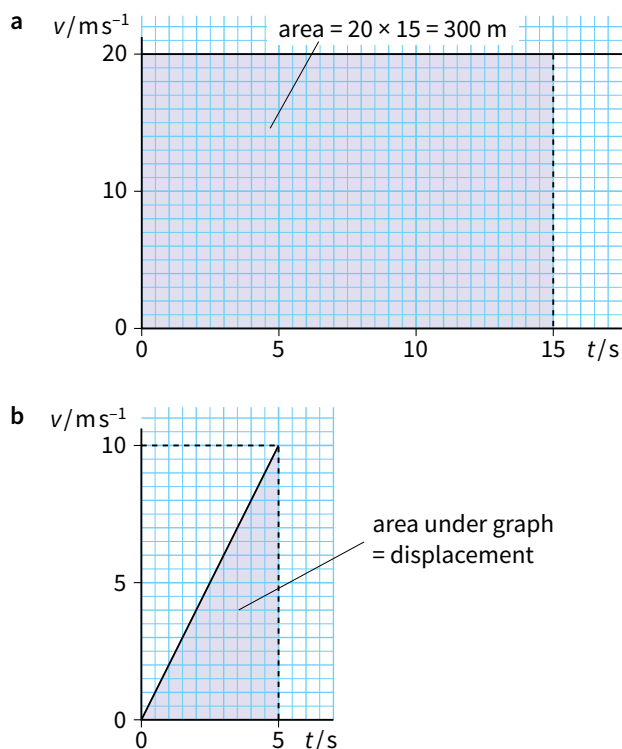


Figure 2.6 The area under the velocity–time graph is equal to the displacement of the object.

(Take care when counting squares: it is easiest when the sides of the squares stand for one unit. Check the axes, as the sides may represent 2 units, or 5 units, or some other number.)

Measuring velocity and acceleration

In a car crash, the occupants of the car may undergo a very rapid deceleration. This can cause them serious injury, but can be avoided if an air-bag is inflated within a fraction of a second. [Figure 2.7](#) shows the tiny accelerometer at the heart of the system, which detects large accelerations and decelerations.

The acceleration sensor consists of two rows of interlocking teeth. In the event of a crash, these move relative to one another, and this generates a voltage which triggers the release of the air-bag.

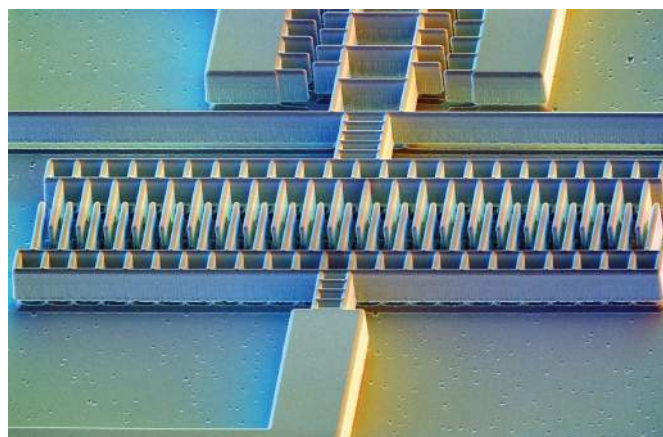


Figure 2.7 A micro-mechanical acceleration sensor is used to detect sudden accelerations and decelerations as a vehicle travels along the road. This electron microscope image shows the device magnified about 1000 times.

At the top of the photograph, you can see a second sensor which detects sideways accelerations. This is important in the case of a side impact.

These sensors can also be used to detect when a car swerves or skids, perhaps on an icy road. In this case, they activate the car's stability-control systems.

Determining velocity and acceleration in the laboratory

In [Chapter 1](#), we looked at ways of finding the velocity of a trolley moving in a straight line. These involved measuring distance and time, and deducing velocity. [Box 2.1](#) below shows how these techniques can be extended to find the acceleration of a trolley.

QUESTIONS

- 4 A lorry driver is travelling at the speed limit on a motorway. Ahead, he sees hazard lights and gradually slows down. He sees that an accident has occurred, and brakes suddenly to a halt. Sketch a velocity–time graph to represent the motion of this lorry.
- 5 [Table 2.1](#) shows how the velocity of a motorcyclist changed during a speed trial along a straight road.
 - a Draw a velocity–time graph for this motion.
 - b From the table, deduce the motorcyclist's acceleration during the first 10 s.
 - c Check your answer by finding the gradient of the graph during the first 10 s.
 - d Determine the motorcyclist's acceleration during the last 15 s.
 - e Use the graph to find the total distance travelled during the speed trial.

Velocity / m s^{-1}	0	15	30	30	20	10	0
Time / s	0	5	10	15	20	25	30

Table 2.1 Data for a motorcyclist.

BOX 2.1: Laboratory measurements of acceleration

Measurements using light gates

The computer records the time for the first ‘interrupt’ section of the card to pass through the light beam of the light gate (Figure 2.8). Given the length of the interrupt, it can work out the trolley’s initial velocity u . This is repeated for the second interrupt to give final velocity v . The computer also records the time interval $t_3 - t_1$ between these two velocity measurements. Now it can calculate the acceleration a as shown below:

$$u = \frac{l_1}{t_2 - t_1}$$

(l_1 = length of first section of the interrupt card)

and

$$v = \frac{l_2}{t_4 - t_3}$$

(l_2 = length of second section of the interrupt card)

Therefore:

$$a = \frac{\text{change in velocity}}{\text{time taken}} = \frac{v - u}{t_3 - t_1}$$

(Note that this calculation gives only an approximate value for a . This is because u and v are **average** speeds over a period of time; for an accurate answer we would need to know the speeds at times t_1 and t_3 .)

Sometimes two light gates are used with a card of length l . The computer can still record the times as shown above and calculate the acceleration in the same way, with $l_1 = l_2 = l$.

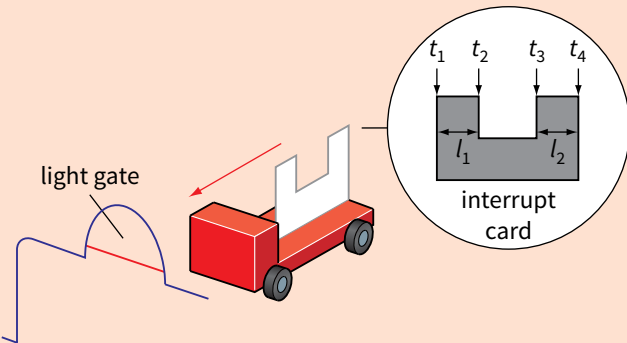


Figure 2.8 Determining acceleration using a single light gate.

Measurements using a ticker-timer

The practical arrangement is the same as for measuring velocity. Now we have to think about how to interpret the tape produced by an accelerating trolley (Figure 2.9).

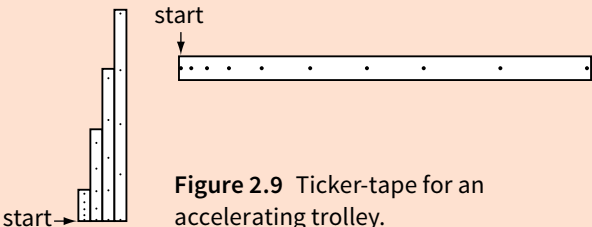


Figure 2.9 Ticker-tape for an accelerating trolley.

The tape is divided into sections, as before, every five dots. Remember that the time interval between adjacent dots is 0.02 s. Each section represents 0.10 s.

By placing the sections of tape side by side, you can picture the velocity–time graph.

The length of each section gives the trolley’s displacement in 0.10 s, from which the average velocity during this time can be found. This can be repeated for each section of the tape, and a velocity–time graph drawn. The gradient of this graph is equal to the acceleration. Table 2.2 and Figure 2.10 show some typical results.

The acceleration is calculated to be:

$$a = \frac{\Delta v}{\Delta t} = \frac{0.93}{0.20} \approx 4.7 \text{ m s}^{-2}$$

Section of tape	Time at start / s	Time interval / s	Length of section / cm	Velocity / m s ⁻¹
1	0.0	0.10	2.3	0.23
2	0.10	0.10	7.0	0.70
3	0.20	0.10	11.6	1.16

Table 2.2 Data for Figure 2.10.

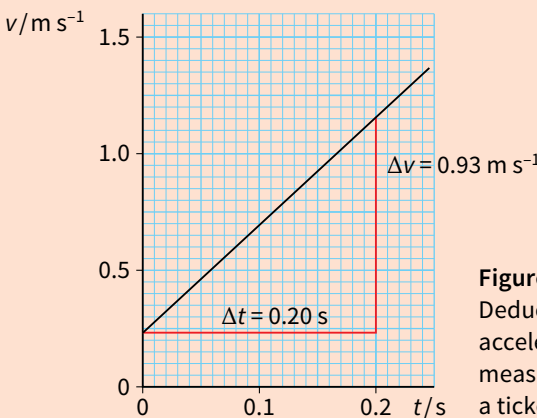


Figure 2.10 Deducing acceleration from measurements of a ticker-tape.

BOX 2.1: Laboratory measurements of acceleration (continued)**Measurements using a motion sensor**

The computer software which handles the data provided by the motion sensor can calculate the acceleration of a trolley. However, because it deduces velocity from measurements of position, and then calculates acceleration from values of velocity, its precision is relatively poor.

QUESTIONS

- 6 Sketch a section of ticker-tape for a trolley which travels at a steady velocity and which then decelerates.
- 7 **Figure 2.11** shows the dimensions of an interrupt card, together with the times recorded as it passed through a light gate. Use these measurements to calculate the acceleration of the card. (Follow the steps outlined on page 19.)

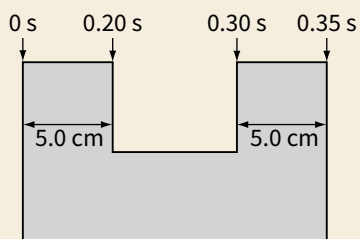


Figure 2.11 For Question 7.

- 8 Two adjacent five-dot sections of a ticker-tape measure 10 cm and 16 cm, respectively. The interval between dots is 0.02 s. Deduce the acceleration of the trolley which produced the tape.

The equations of motion

As a space rocket rises from the ground, its velocity steadily increases. It is accelerating (**Figure 2.12**).

Eventually it will reach a speed of several kilometres per second. Any astronauts aboard find themselves pushed back into their seats while the rocket is accelerating.

The engineers who planned the mission must be able to calculate how fast the rocket will be travelling and where it will be at any point in its journey. They have sophisticated computers to do this, using more elaborate versions of the equations given below.



Figure 2.12 A rocket accelerates as it lifts off from the ground.

There is a set of equations which allows us to calculate the quantities involved when an object is moving with a constant acceleration. The quantities we are concerned with are:

s	displacement	a	acceleration
u	initial velocity	t	time taken
v	final velocity		

Here are the four **equations of motion**.

equation 1:	$v = u + at$
equation 2:	$s = \frac{(u+v)}{2} \times t$
equation 3:	$s = ut + \frac{1}{2}at^2$
equation 4:	$v^2 = u^2 + 2as$

Take care using these equations. They can only be used:

- for motion in a straight line
- for an object with constant acceleration.

To get a feel for how to use these equations, we will consider some worked examples. In each example, we will follow the same procedure:

- Step 1** We write down the quantities which we know, and the quantity we want to find.
- Step 2** Then we choose the equation which links these quantities, and substitute in the values.
- Step 3** Finally, we calculate the unknown quantity.

We will look at where these equations come from in the next section.

WORKED EXAMPLES

- 4 The rocket shown in Figure 2.12 lifts off from rest with an acceleration of 20 m s^{-2} . Calculate its velocity after 50 s.

Step 1 What we know: $u = 0 \text{ m s}^{-1}$
 $a = 20 \text{ m s}^{-2}$
 $t = 50 \text{ s}$

and what we want to know: $v = ?$

Step 2 The equation linking u , a , t and v is equation 1:
 $v = u + at$

Substituting gives:

$$v = 0 + (20 \times 50)$$

Step 3 Calculation then gives:

$$v = 1000 \text{ m s}^{-1}$$

So the rocket will be travelling at 1000 m s^{-1} after 50 s. This makes sense, since its velocity increases by 20 m s^{-1} every second, for 50 s.

You could use the same equation to work out how long the rocket would take to reach a velocity of 2000 m s^{-1} , or the acceleration it must have to reach a speed of 1000 m s^{-1} in 40 s, and so on.

- 5 The car shown in Figure 2.13 is travelling along a straight road at 8.0 m s^{-1} . It accelerates at 1.0 m s^{-2} for a distance of 18 m. How fast is it then travelling?

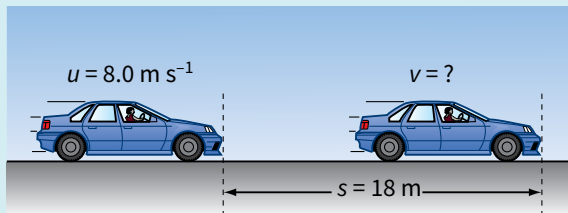


Figure 2.13 For Worked example 5. This car accelerates for a short distance as it travels along the road.

In this case, we will have to use a different equation, because we know the distance during which the car accelerates, not the time.

Step 1 What we know: $u = 8.0 \text{ m s}^{-1}$
 $a = 1.0 \text{ m s}^{-2}$
 $s = 18 \text{ m}$

and what we want to know: $v = ?$

Step 2 The equation we need is equation 4:
 $v^2 = u^2 + 2as$

Substituting gives:

$$v^2 = 8.0^2 + (2 \times 1.0 \times 18)$$

Step 3 Calculation then gives:

$$v^2 = 64 + 36 = 100 \text{ m}^2 \text{ s}^{-2}$$

$$v = 10 \text{ m s}^{-1}$$

So the car will be travelling at 10 m s^{-1} when it stops accelerating.

(You may find it easier to carry out these calculations without including the units of quantities when you substitute in the equation. However, including the units can help to ensure that you end up with the correct units for the final answer.)

- 6 A train (Figure 2.14) travelling at 20 m s^{-1} accelerates at 0.50 m s^{-2} for 30 s. Calculate the distance travelled by the train in this time.

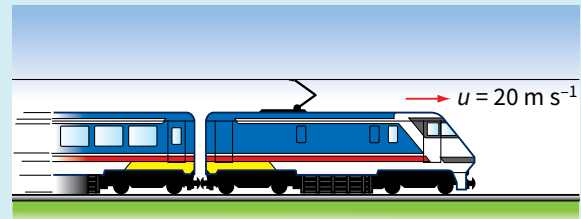


Figure 2.14 For Worked example 6. This train accelerates for 30 s.

Step 1 What we know: $u = 20 \text{ m s}^{-1}$
 $t = 30 \text{ s}$
 $a = 0.50 \text{ m s}^{-2}$

and what we want to know: $s = ?$

Step 2 The equation we need is equation 3:

$$s = ut + \frac{1}{2}at^2$$

Substituting gives:

$$s = (20 \times 30) + \frac{1}{2} \times 0.5 \times (30)^2$$

Step 3 Calculation then gives:

$$s = 600 + 225 = 825 \text{ m}$$

So the train will travel 825 m while it is accelerating.

WORKED EXAMPLES (continued)

- 7** The cyclist in Figure 2.15 is travelling at 15 m s^{-1} . She brakes so that she doesn't collide with the wall. Calculate the magnitude of her deceleration.

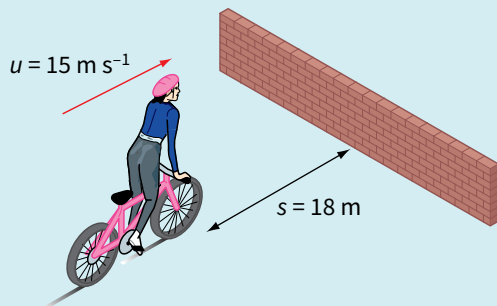


Figure 2.15 For Worked example 7. The cyclist brakes to stop herself colliding with the wall.

This example shows that it is sometimes necessary to rearrange an equation, to make the unknown quantity its subject. It is easiest to do this before substituting in the values.

Step 1 What we know: $u = 15 \text{ m s}^{-1}$
 $v = 0 \text{ m s}^{-1}$
 $s = 18 \text{ m}$

and what we want to know: $a = ?$

Step 2 The equation we need is equation 4:

$$v^2 = u^2 + 2as$$

Rearranging gives:

$$a = \frac{v^2 - u^2}{2s}$$

$$a = \frac{0^2 - 15^2}{2 \times 18} = \frac{-225}{36}$$

Step 3 Calculation then gives:

$$a = -6.25 \text{ m s}^{-2} \approx -6.3 \text{ m s}^{-2}$$

So the cyclist will have to brake hard to achieve a deceleration of magnitude 6.3 m s^{-2} . The minus sign shows that her acceleration is negative, i.e. a deceleration.

QUESTIONS

- 9** A car is initially stationary. It has a constant acceleration of 2.0 m s^{-2} .
- Calculate the velocity of the car after 10 s.
 - Calculate the distance travelled by the car at the end of 10 s.
 - Calculate the time taken by the car to reach a velocity of 24 m s^{-1} .
- 10** A train accelerates steadily from 4.0 m s^{-1} to 20 m s^{-1} in 100 s.
- Calculate the acceleration of the train.
 - From its initial and final velocities, calculate the average velocity of the train.
 - Calculate the distance travelled by the train in this time of 100 s.
- 11** A car is moving at 8.0 m s^{-1} . The driver makes it accelerate at 1.0 m s^{-2} for a distance of 18 m. What is the final velocity of the car?

Deriving the equations of motion

On the previous pages, we have seen how to make use of the equations of motion. But where do these equations come from? They arise from the definitions of velocity and acceleration.

We can find the first two equations from the velocity–time graph shown in Figure 2.16. The graph represents the motion of an object. Its initial velocity is u . After time t , its final velocity is v .

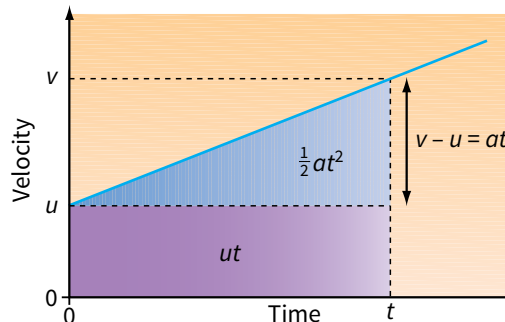


Figure 2.16 This graph shows the variation of velocity of an object with time. The object has constant acceleration.

Equation 1

The graph of Figure 2.16 is a straight line, therefore the object's acceleration a is constant. The gradient (slope) of the line is equal to acceleration.

The acceleration is defined as:

$$a = \frac{(v - u)}{t}$$

which is the gradient of the line. Rearranging this gives the first equation of motion:

$$v = u + at \quad (\text{equation 1})$$

Equation 2

Displacement is given by the area under the velocity–time graph. Figure 2.17 shows that the object's average velocity is half-way between u and v . So the object's average velocity, calculated by averaging its initial and final velocities, is given by:

$$\frac{(u + v)}{2}$$

The object's displacement is the shaded area in Figure 2.17. This is a rectangle, and so we have:

displacement = average velocity $\times$ time taken

and hence:

$$s = \frac{(u + v)}{2} \times t \quad (\text{equation 2})$$

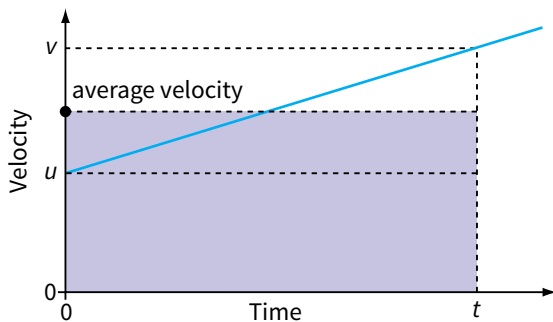


Figure 2.17 The average velocity is half-way between u and v .

Equation 3

From equations 1 and 2, we can derive equation 3:

$$v = u + at \quad (\text{equation 1})$$

$$s = \frac{(u + v)}{2} \times t \quad (\text{equation 2})$$

Substituting v from equation 1 gives:

$$s = \frac{(u + u + at)}{2} \times t$$

$$s = \frac{2ut}{2} + \frac{at^2}{2}$$

So

$$s = ut + \frac{1}{2}at^2 \quad (\text{equation 3})$$

Looking at Figure 2.16, you can see that the two terms on the right of the equation correspond to the areas of the rectangle and the triangle which make up the area under the graph. Of course, this is the same area as the rectangle in Figure 2.17.

Equation 4

Equation 4 is also derived from equations 1 and 2:

$$v = u + at \quad (\text{equation 1})$$

$$s = \frac{(u + v)}{2} \times t \quad (\text{equation 2})$$

Substituting for time t from equation 1 gives:

$$s = \frac{(u + v)}{2} + \frac{(v + u)}{a} \quad (\text{equation 2})$$

Rearranging this gives:

$$\begin{aligned} 2as &= (u + v)(v - u) \\ &= v^2 - u^2 \end{aligned}$$

or simply:

$$v^2 = u^2 + 2as \quad (\text{equation 4})$$

Investigating road traffic accidents

The police frequently have to investigate road traffic accidents. They make use of many aspects of physics, including the equations of motion. The next two questions will help you to apply what you have learned to situations where police investigators have used evidence from skid marks on the road.

QUESTIONS

- 12** Trials on the surface of a new road show that, when a car skids to a halt, its acceleration is -7.0 m s^{-2} . Estimate the skid-to-stop distance of a car travelling at a speed limit of 30 m s^{-1} (approx. 110 km h^{-1} or 70 mph).
- 13** At the scene of an accident on a country road, police find skid marks stretching for 50 m. Tests on the road surface show that a skidding car decelerates at 6.5 m s^{-2} . Was the car which skidded exceeding the speed limit of 25 m s^{-1} (90 km h^{-1}) on this road?

Uniform and non-uniform acceleration

It is important to note that the equations of motion only apply to an object which is moving with a constant acceleration. If the acceleration a was changing, you wouldn't know what value to put in the equations. Constant acceleration is often referred to as uniform acceleration.

The velocity–time graph in Figure 2.18 shows **non-uniform** acceleration. It is not a straight line; its gradient is changing (in this case, decreasing).

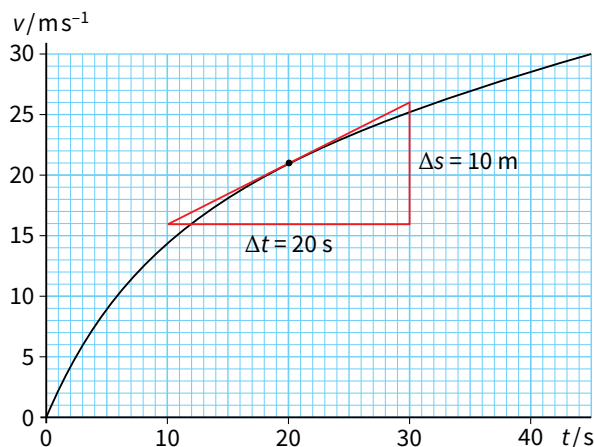


Figure 2.18 This curved velocity–time graph cannot be analysed using the equations of motion.

The acceleration at any instant in time is given by the gradient of the velocity–time graph. The triangle in Figure 2.18 shows how to find the acceleration at $t = 20$ seconds:

- At the time of interest, mark a point on the graph.
- Draw a **tangent** to the curve at that point.
- Make a large right-angled triangle, and use it to find the gradient.

You can find the change in displacement of the body as it accelerates by determining the area under the velocity–time graph.

To find the displacement of the object in Figure 2.18 between $t = 0$ and 20 s, the most straightforward, but lengthy, method is just to count the number of small squares.

In this case up to $t = 20$ s, there are about 250 small squares. This is tedious to count but you can save yourself a lot of time by drawing a line from the origin to the point at 20 s. The area of the triangle is easy to find (200 small squares) and then you only have to count the number of small squares between the line you have drawn and the curve on the graph (about 50 squares)

In this case each square is 1 m s^{-1} on the y-axis by 1 s on the x-axis, so the area of each square is $1 \times 1 = 1 \text{ m}$ and the displacement is 250 m. In other cases note carefully the value of each side of the square you have chosen.

QUESTIONS

- 14 The graph in Figure 2.19 represents the motion of an object moving with varying acceleration. Lay your ruler on the diagram so that it is tangential to the graph at point P.

- a What are the values of time and velocity at this point?
- b Estimate the object's acceleration at this point.

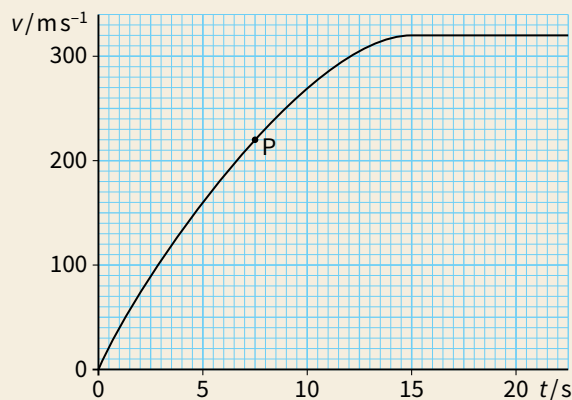


Figure 2.19 For Question 14.

- 15 The velocity–time graph (Figure 2.20) represents the motion of a car along a straight road for a period of 30 s.

- a Describe the motion of the car.
- b From the graph, determine the car's initial and final velocities over the time of 30 s.
- c Determine the acceleration of the car.
- d By calculating the area under the graph, determine the displacement of the car.
- e Check your answer to part d by calculating the car's displacement using $s = ut + \frac{1}{2}at^2$.

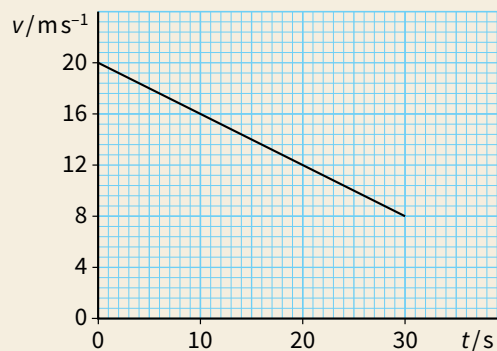


Figure 2.20 For Question 15.

Acceleration caused by gravity

If you drop a ball or stone, it falls to the ground. [Figure 2.21](#), based on a multiframe photograph, shows the ball at equal intervals of time. You can see that the ball's velocity increases as it falls because the spaces between the images of the ball increase steadily. The ball is accelerating.

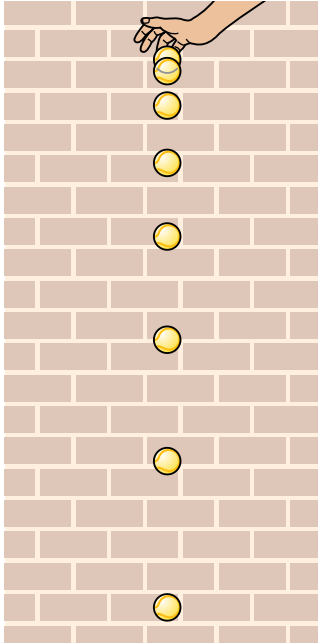


Figure 2.21 This diagram of a falling ball, based on a multiframe photo, clearly shows that the ball's velocity increases as it falls.

A multiframe photograph is useful to demonstrate that the ball accelerates as it falls. Usually, objects fall too quickly for our eyes to be able to observe them speeding up. It is easy to imagine that the ball moves quickly as soon as you let it go, and falls at a steady speed to the ground.

[Figure 2.21](#) shows that this is not the case.

If we measure the acceleration of a freely falling object on the surface of the Earth, we find a value of about 9.81 m s^{-2} . This is known as the **acceleration of free fall**, and is given the symbol g :

$$\text{acceleration of free fall, } g = 9.81 \text{ m s}^{-2}$$

The value of g depends on where you are on the Earth's surface, but we usually take $g = 9.81 \text{ m s}^{-2}$.

If we drop an object, its initial velocity $u = 0$. How far will it fall in time t ? Substituting in $s = ut + \frac{1}{2}at^2$ gives displacement s :

$$\begin{aligned} s &= \frac{1}{2} \times 9.81 \times t^2 \\ &= 4.9 \times t^2 \end{aligned}$$

Hence, by timing a falling object, we can determine g .

QUESTIONS

- 16** If you drop a stone from the edge of a cliff, its initial velocity $u = 0$, and it falls with acceleration $g = 9.81 \text{ m s}^{-2}$. You can calculate the distance s it falls in a given time t using an equation of motion.
- Copy and complete [Table 2.3](#), which shows how s depends on t .
 - Draw a graph of s against t .
 - Use your graph to find the distance fallen by the stone in 2.5 s.
 - Use your graph to find how long it will take the stone to fall to the bottom of a cliff 40 m high. Check your answer using the equations of motion.

Time / s	0	1.0	2.0	3.0	4.0
Displacement / m	0	4.9			

Table 2.3 Time (t) and displacement (s) data for Question 16.

- 17** An egg falls off a table. The floor is 0.8 m from the table-top.
- Calculate the time taken to reach the ground.
 - Calculate the velocity of impact with the ground.

Determining g

One way to measure the acceleration of free fall g would be to try bungee-jumping ([Figure 2.22](#)). You would need to carry a stopwatch, and measure the time between jumping from the platform and the moment when the elastic rope begins to slow your fall. If you knew the length of the unstretched rope, you could calculate g .

There are easier methods for finding g which can be used in the laboratory. These are described in [Box 2.2](#).



Figure 2.22 A bungee-jumper falls with initial acceleration g .

BOX 2.2: Laboratory measurements of g **Measuring g using an electronic timer**

In this method, a steel ball-bearing is held by an electromagnet (Figure 2.23). When the current to the magnet is switched off, the ball begins to fall and an electronic timer starts. The ball falls through a trapdoor, and this breaks a circuit to stop the timer. This tells us the time taken for the ball to fall from rest through the distance h between the bottom of the ball and the trapdoor.

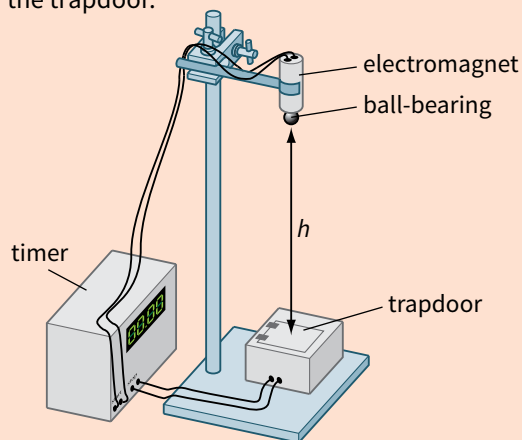


Figure 2.23 The timer records the time for the ball to fall through the distance h .

Here is how we can use one of the equations of motion to find g :

$$\text{displacement } s = h$$

$$\text{time taken} = t$$

$$\text{initial velocity } u = 0$$

$$\text{acceleration } a = g$$

Substituting in $s = ut + \frac{1}{2}at^2$ gives:

$$h = \frac{1}{2}gt^2$$

and for any values of h and t we can calculate a value for g .

A more satisfactory procedure is to take measurements of t for several different values of h . The height of the ball bearing above the trapdoor is varied systematically, and the time of fall measured several times to calculate an average for each height. Table 2.4 and Figure 2.24 show some typical results. We can deduce g from the gradient of the graph of h against t^2 . The equation for a straight line through the origin is:

$$y = mx$$

In our experiment we have:

$$\begin{array}{c} h = \frac{1}{2}gt^2 \\ y = mx \end{array}$$

h/m	t/s	t^2/s^2
0.27	0.25	0.063
0.39	0.30	0.090
0.56	0.36	0.130
0.70	0.41	0.168
0.90	0.46	0.212

Table 2.4 Data for Figure 2.24. These are mean values.

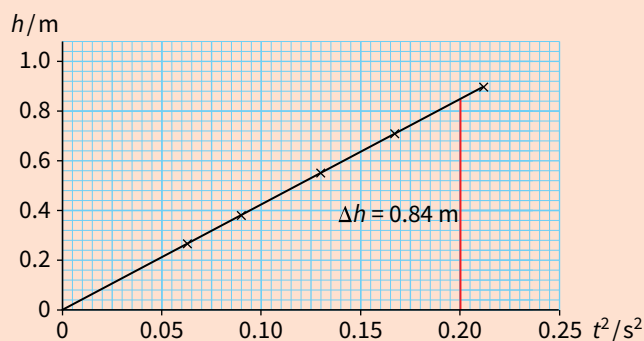


Figure 2.24 The acceleration of free fall can be determined from the gradient.

The gradient of the straight line of a graph of h against t^2 is equal to $\frac{g}{2}$. Therefore:

$$\text{gradient} = \frac{g}{2} = \frac{0.84}{0.20} = 4.2$$

$$g = 4.2 \times 2 = 8.4 \text{ m s}^{-2}$$

Sources of uncertainty

The electromagnet may retain some magnetism when it is switched off, and this may tend to slow the ball's fall. Consequently, the time t recorded by the timer may be longer than if the ball were to fall completely freely. From $h = \frac{1}{2}gt^2$, it follows that, if t is too great, the experimental value of g will be too small. This is an example of a systematic error – all the results are systematically distorted so that they are too great (or too small) as a consequence of the experimental design.

Measuring the height h is awkward. You can probably only find the value of h to within $\pm 1 \text{ mm}$ at best. So there is a random error in the value of h , and this will result in a slight scatter of the points on the graph, and a degree of uncertainty in the final value of g . For more about errors, see P1: Practical skills for AS.

BOX 2.2: Laboratory measurements of g (continued)**Measuring g using a ticker-timer**

Figure 2.25 shows a weight falling. As it falls, it pulls a tape through a ticker-timer. The spacing of the dots on the tape increases steadily, showing that the weight is accelerating. You can analyse the tape to find the acceleration, as discussed on page 19.

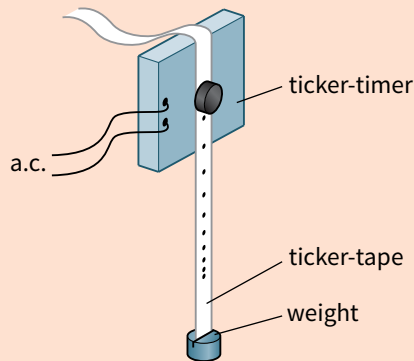


Figure 2.25 A falling weight pulls a tape through a ticker-timer.

This is not a very satisfactory method of measuring g . The main problem arises from friction between the tape and the ticker-timer. This slows the fall of the weight and so its acceleration is less than g . (This is another example of a systematic error.)

The effect of friction is less of a problem for a large weight, which falls more freely. If measurements are made for increasing weights, the value of acceleration gets closer and closer to the true value of g .

Measuring g using a light gate

Figure 2.26 shows how a weight can be attached to a card 'interrupt'. The card is designed to break the light beam twice as the weight falls. The computer can then calculate the velocity of the weight twice as it falls, and hence find its acceleration:

$$\text{initial velocity } u = \frac{x}{t_2 - t_1}$$

$$\text{final velocity } v = \frac{x}{t_4 - t_3}$$

Therefore:

$$\text{acceleration } a = \frac{v - u}{t_3 - t_1}$$

The weight can be dropped from different heights above the light gate. This allows you to find out whether its acceleration is the same at different points in its fall. This is an advantage over Method 1, which can only measure the acceleration from a stationary start.

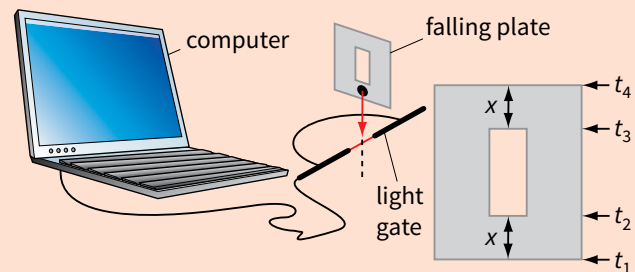


Figure 2.26 The weight accelerates as it falls. The upper section of the card falls more quickly through the light gate.

WORKED EXAMPLE

- 8** To get a rough value for g , a student dropped a stone from the top of a cliff. A second student timed the stone's fall using a stopwatch. Here are their results:

estimated height of cliff = 30 m

time of fall = 2.6 s

Use the results to estimate a value for g .

Step 1 Calculate the average speed of the stone:

$$\text{average speed of stone during fall} = \frac{30}{2.6} = 11.5 \text{ m s}^{-1}$$

Step 2 Find the values of v and u :

$$\text{final speed } v = 2 \times 11.5 \text{ m s}^{-1} = 23.0 \text{ m s}^{-1}$$

$$\text{initial speed } u = 0 \text{ m s}^{-1}$$

Step 3 Substitute these values into the equation for acceleration:

$$a = \frac{v - u}{t} = \frac{23.0}{2.6} = 8.8 \text{ m s}^{-2}$$

Note that you can reach the same result more directly using $s = ut + \frac{1}{2}at^2$, but you may find it easier to follow what is going on using the method given here. We should briefly consider why the answer is less than the expected value of $g = 9.81 \text{ m s}^{-2}$. It might be that the cliff was higher than the student's estimate. The timer may not have been accurate in switching the stopwatch on and off. There will have been air resistance which slowed the stone's fall.

QUESTIONS

- 18 A steel ball falls from rest through a height of 2.10 m. An electronic timer records a time of 0.67 s for the fall.
- Calculate the average acceleration of the ball as it falls.
 - Suggest reasons why the answer is not exactly 9.81 m s^{-2} .
- 19 In an experiment to determine the acceleration due to gravity, a ball was timed electronically as it fell from rest through a height h . The times t shown in Table 2.5 were obtained.
- Plot a graph of h against t^2 .
 - From the graph, determine the acceleration of free fall, g .
 - Comment on your answer.

Height / m	0.70	1.03	1.25	1.60	1.99
Time / s	0.99	1.13	1.28	1.42	1.60

Table 2.5 Height (h) and time (t) data for Question 19.

- 20 In Chapter 1, we looked at how to use a motion sensor to measure the speed and position of a moving object. Suggest how a motion sensor could be used to determine g .

Motion in two dimensions – projectiles

A curved trajectory

A multiframe photograph can reveal details of the path, or trajectory, of a projectile. Figure 2.27 shows the trajectories of a projectile – a bouncing ball. Once the ball has left the child's hand and is moving through the air, the only force acting on it is its weight.

The ball has been thrown at an angle to the horizontal. It speeds up as it falls – you can see that the images of the ball become further and further apart. At the same time, it moves steadily to the right. You can see this from the even spacing of the images across the picture. The ball's path has a mathematical shape known as a **parabola**. After it bounces, the ball is moving more slowly. It slows down, or decelerates, as it rises – the images get closer and closer together.

We interpret this picture as follows. The vertical motion of the ball is affected by the force of gravity, that is, its weight. When it rises it has a vertical deceleration



Figure 2.27 A bouncing ball is an example of a projectile. This multiframe photograph shows details of its motion which would escape the eye of an observer.

of magnitude g , which slows it down, and when it falls it has an acceleration of g , which speeds it up. The ball's horizontal motion is unaffected by gravity. In the absence of air resistance, the ball has a constant velocity in the horizontal direction. We can treat the ball's vertical and horizontal motions separately, because they are independent of one another.

Components of a vector

In order to understand how to treat the velocity in the vertical and horizontal directions separately we start by considering a constant velocity.

If an aeroplane has a constant velocity v at an angle θ as shown in Figure 2.28, then we say that this velocity has two effects or **components**, v_N in a northerly direction and v_E in an easterly direction. These two components of velocity add up to make the actual velocity v .

This process of taking a velocity and determining its effect along another direction is known as **resolving** the velocity along a different direction. In effect splitting the velocity into two components at right angles is the reverse

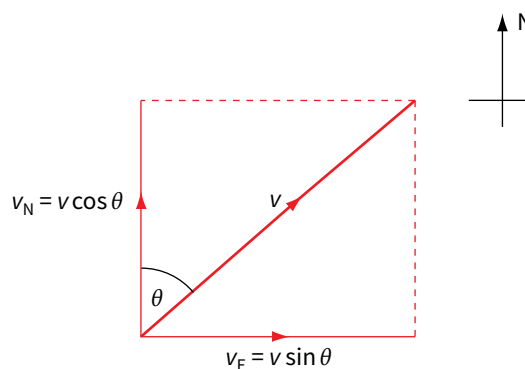


Figure 2.28 Components of a velocity. The component due north is $v_N = v \cos \theta$ and the component due east is $v_E = v \sin \theta$.

of adding together two vectors – it is splitting one vector into two vectors along convenient directions.

To find the component of any vector (e.g. displacement, velocity, acceleration) in a particular direction, we can use the following strategy:

Step 1 Find the angle θ between the vector and the direction of interest.

Step 2 Multiply the vector by the cosine of the angle θ .

So the component of an object's velocity v at angle θ to v is equal to $v \cos \theta$ (Figure 2.28).

QUESTION

- 21** Find the x - and y -components of each of the vectors shown in Figure 2.29. (You will need to use a protractor to measure angles from the diagram.)

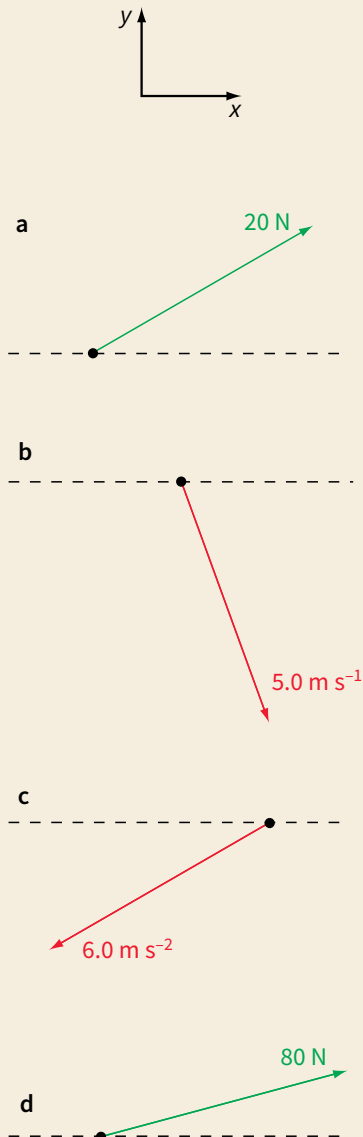


Figure 2.29
The vectors for
Question 21.

Understanding projectiles

We will first consider the simple case of a projectile thrown straight up in the air, so that it moves vertically. Then we will look at projectiles which move horizontally and vertically at the same time.

Up and down

A stone is thrown upwards with an initial velocity of 20 m s^{-1} . Figure 2.30 shows the situation.

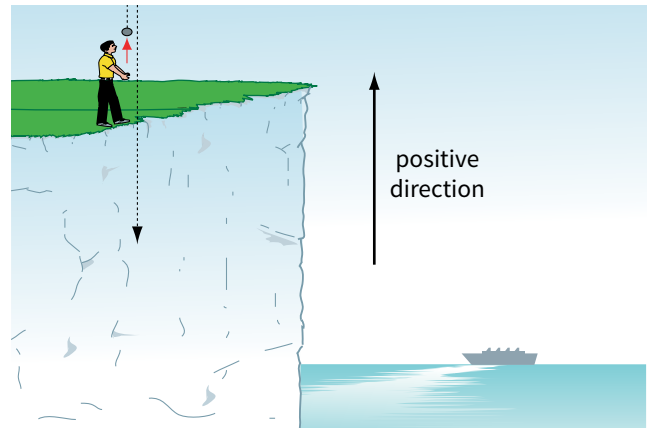


Figure 2.30 Standing at the edge of the cliff, you throw a stone vertically upwards. The height of the cliff is 25 m.

It is important to use a consistent sign convention here. We will take upwards as positive, and downwards as negative. So the stone's initial velocity is positive, but its acceleration g is negative. We can solve various problems about the stone's motion by using the equations of motion.

How high?

How high will the stone rise above ground level of the cliff?

As the stone rises upwards, it moves more and more slowly – it decelerates, because of the force of gravity. At its highest point, the stone's velocity is zero. So the quantities we know are:

$$\text{initial velocity} = u = 20 \text{ m s}^{-1}$$

$$\text{final velocity} = v = 0 \text{ m s}^{-1}$$

$$\text{acceleration} = a = -9.81 \text{ m s}^{-2}$$

$$\text{displacement} = s = ?$$

The relevant equation of motion is $v^2 = u^2 + 2as$.

Substituting values gives:

$$0^2 = 20^2 + 2 \times (-9.81) \times s$$

$$0 = 400 - 19.62s$$

$$s = \frac{400}{19.62} = 20.4 \text{ m} \approx 20 \text{ m}$$

The stone rises 20 m upwards, before it starts to fall again.

How long?

How long will it take from leaving your hand for the stone to fall back to the clifftop?

When the stone returns to the point from which it was thrown, its displacement s is zero. So:

$s = 0 \quad u = 20 \text{ m s}^{-1} \quad a = -9.81 \text{ m s}^{-2} \quad t = ?$

Substituting in $s = ut + \frac{1}{2}at^2$ gives:

$$0 = 20t + \frac{1}{2}(-9.81) \times t^2$$
$$= 20t - 4.905t^2 = (20 - 4.905t) \times t$$

There are two possible solutions to this:

- $t = 0 \text{ s}$, i.e. the stone had zero displacement at the instant it was thrown
- $t = 4.1 \text{ s}$, i.e. the stone returned to zero displacement after 4.1 s, which is the answer we are interested in.

Falling further

The height of the cliff is 25 m. How long will it take the stone to reach the foot of the cliff?

This is similar to the last example, but now the stone's final displacement is 25 m below its starting point. By our sign convention, this is a negative displacement, and $s = -25 \text{ m}$.

also shown in Figure 2.31. Study the table and the graph. You should notice the following:

- The horizontal distance increases steadily. This is because the ball's horizontal motion is unaffected by the force of gravity. It travels at a steady velocity horizontally so we can use $v = \frac{s}{t}$.
- The vertical distances do not show the same pattern. The ball is accelerating downwards so we must use the equations of motion. (These figures have been calculated using $g = 9.81 \text{ m s}^{-2}$.)

Time / s	Horizontal distance / m	Vertical distance / m
0.00	0.00	0.000
0.04	0.10	0.008
0.08	0.20	0.031
0.12	0.30	0.071
0.16	0.40	0.126
0.20	0.50	0.196
0.24	0.60	0.283
0.28	0.70	0.385

Table 2.7 Data for the example of a moving ball, as shown in Figure 2.31.

You can calculate the distance s fallen using the equation of motion $s = ut + \frac{1}{2}at^2$. (The initial vertical velocity $u = 0$.)

The horizontal distance is calculated using:

$\text{horizontal distance} = 2.5 \times t$

The vertical distance is calculated using:

$\text{vertical distance} = \frac{1}{2} \times 9.81 \times t^2$

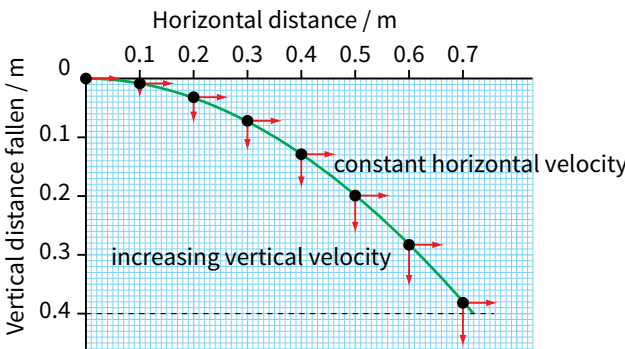


Figure 2.31 This sketch shows the path of the ball projected horizontally. The arrows represent the horizontal and vertical components of its velocity.

QUESTIONS

- 22 In the example above (Falling further), calculate the time it will take for the stone to reach the foot of the cliff.
- 23 A ball is fired upwards with an initial velocity of 30 m s^{-1} . Table 2.6 shows how the ball's velocity changes. (Take $g = 9.81 \text{ m s}^{-2}$.)
- a Copy and complete the table.
 - b Draw a graph to represent the data.
 - c Use your graph to deduce how long the ball took to reach its highest point.

Velocity / m s^{-1}	30	20.19				
Time / s	0	1.0	2.0	3.0	4.0	5.0

Table 2.6 For Question 23.

Vertical and horizontal at the same time

Here is an example to illustrate what happens when an object travels vertically and horizontally at the same time.

In a toy, a ball-bearing is fired horizontally from a point 0.4 m above the ground. Its initial velocity is 2.5 m s^{-1} . Its positions at equal intervals of time have been calculated and are shown in Table 2.7. These results are

WORKED EXAMPLES

- 9** A stone is thrown horizontally with a velocity of 12 m s^{-1} from the top of a vertical cliff.

Calculate how long the stone takes to reach the ground 40 m below and how far the stone lands from the base of the cliff.

Step 1 Consider the ball's vertical motion. It has zero initial speed vertically and travels 40 m with acceleration 9.81 m s^{-2} in the same direction.

$$s = ut + \frac{1}{2}at^2$$

$$40 = 0 + \frac{1}{2} \times 9.81 \times t^2$$

Thus $t = 2.86 \text{ s}$.

Step 2 Consider the ball's horizontal motion. The ball travels with a constant horizontal velocity, 12 m s^{-1} , as long as there is no air resistance.

$$\text{distance travelled} = u \times t = 12 \times 2.86 = 34.3 \text{ m}$$

Hint: You may find it easier to summarise the information like this:

$$\text{vertically} \quad s = 40 \quad u = 0 \quad a = 9.81 \quad t = ? \quad v = ?$$

$$\text{horizontally} \quad u = 12 \quad v = 12 \quad a = 0 \quad t = ? \quad s = ?$$

- 10** A ball is thrown with an initial velocity of 20 m s^{-1} at an angle of 30° to the horizontal (Figure 2.32). Calculate the horizontal distance travelled by the ball (its **range**).

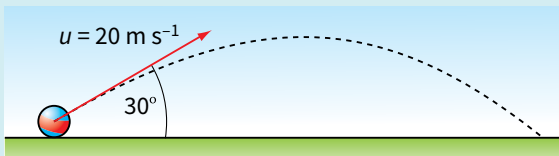


Figure 2.32 Where will the ball land?

Step 1 Split the ball's initial velocity into horizontal and vertical components:

$$\text{initial velocity} = u = 20 \text{ m s}^{-1}$$

$$\text{horizontal component of initial velocity} = u \cos \theta = 20 \times \cos 30^\circ = 17.3 \text{ m s}^{-1}$$

$$\text{vertical component of initial velocity} = u \sin \theta = 20 \times \sin 30^\circ = 10 \text{ m s}^{-1}$$

Step 2 Consider the ball's vertical motion. How long will it take to return to the ground? In other words, when will its displacement return to zero?

$$u = 10 \text{ m s}^{-1} \quad a = -9.81 \text{ m s}^{-2} \quad s = 0 \quad t = ?$$

Using $s = ut + \frac{1}{2}at^2$, we have:

$$0 = 10t - 4.905t^2$$

This gives $t = 0 \text{ s}$ or $t = 2.04 \text{ s}$. So the ball is in the air for 2.04 s.

Step 3 Consider the ball's horizontal motion. How far will it travel horizontally in the 2.04 s before it lands? This is simple to calculate, since it moves with a constant horizontal velocity of 17.3 m s^{-1} .

$$\text{horizontal displacement } s = 17.3 \times 2.04 = 35.3 \text{ m}$$

Hence the horizontal distance travelled by the ball (its range) is about 35 m.

QUESTIONS

- 24** A stone is thrown horizontally from the top of a vertical cliff and lands 4.0 s later at a distance 12.0 m from the base of the cliff. Ignore air resistance.
- Calculate the horizontal speed of the stone.
 - Calculate the height of the cliff.
- 25** A stone is thrown with a velocity of 8 m s^{-1} into the air at an angle of 40° to the horizontal.
- Calculate the vertical component of the velocity.
 - State the value of the vertical component of the velocity when the stone reaches its highest point. Ignore air resistance.
- Use your answers to **a** and **b** to calculate the time the stone takes to reach its highest point.
 - Calculate the horizontal component of the velocity.
 - Use your answers to **c** and **d** to find the horizontal distance travelled by the stone as it climbs to its highest point.
- 26** The range of a projectile is the horizontal distance it travels before it reaches the ground. The greatest range is achieved if the projectile is thrown at 45° to the horizontal.

A ball is thrown with an initial velocity of 40 m s^{-1} . Calculate its greatest possible range when air resistance is considered to be negligible.

Summary

- Acceleration is equal to the rate of change of velocity.
- Acceleration is a vector quantity.
- The gradient of a velocity–time graph is equal to acceleration:

$$a = \frac{\Delta v}{\Delta t}$$

- The area under a velocity–time graph is equal to displacement (or distance travelled).
- The equations of motion (for constant acceleration in a straight line) are:

$$v = u + at \quad s = ut + \frac{1}{2}at^2$$

$$s = \frac{(u + v)}{2}t \quad v^2 = u^2 + 2as$$

- Vectors such as forces can be resolved into components. Components at right angles to one another can be treated independently of one another. For a velocity v at an angle θ to the x -direction, the components are:

$$x\text{-direction: } v \cos \theta$$

$$y\text{-direction: } v \sin \theta$$

- For projectiles, the horizontal and vertical components of velocity can be treated independently. In the absence of air resistance, the horizontal component of velocity is constant while the vertical component of velocity downwards increases at a rate of 9.81 m s^{-2} .

End-of-chapter questions

- 1 A motorway designer can assume that cars approaching a motorway enter a slip road with a velocity of 10 m s^{-1} and reach a velocity of 30 m s^{-1} before joining the motorway. Calculate the minimum length for the slip road, assuming that vehicles have an acceleration of 4.0 m s^{-2} . [4]
- 2 A train is travelling at 50 m s^{-1} when the driver applies the brakes and gives the train a constant deceleration of magnitude 0.50 m s^{-2} for 100 s. Describe what happens to the train. Calculate the distance travelled by the train in 100 s. [7]
- 3 A boy stands on a cliff edge and throws a stone vertically upwards at time $t = 0$. The stone leaves his hand at 20 m s^{-1} . Take the acceleration of the ball as 9.81 m s^{-2} .
 - a Show that the equation for the displacement of the ball is:
 $s = 20t - 4.9t^2$ [2]
 - b What is the height of the stone 2.0 s after release and 6.0 s after release? [3]
 - c When does the stone return to the level of the boy's hand? Assume the boy's hand does not move vertically after the ball is released. [4]

- 4 The graph in Figure 2.33 shows the variation of velocity with time of two cars, A and B, which are travelling in the same direction over a period of time of 40 s. Car A, travelling at a constant velocity of 40 m s^{-1} , overtakes car B at time $t = 0$. In order to catch up with car A, car B immediately accelerates uniformly for 20 s to reach a constant velocity of 50 m s^{-1} . Calculate:

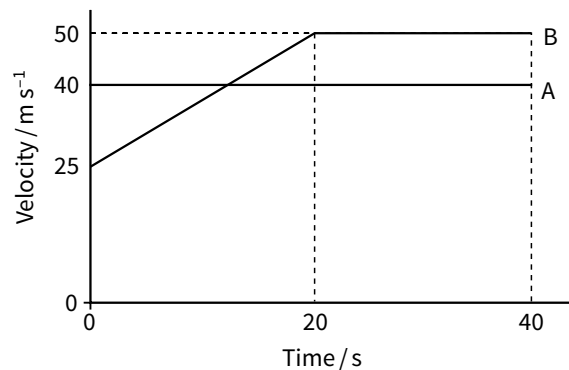


Figure 2.33 Velocity–time graphs for two cars, A and B. For End-of-chapter Question 4.

- a how far A travels during the first 20 s [2]
 - b the acceleration and distance of travel of B during the first 20 s [5]
 - c the additional time taken for B to catch up with A [2]
 - d the distance each car will have then travelled since $t = 0$. [2]
- 5 An athlete competing in the long jump leaves the ground with a velocity of 5.6 m s^{-1} at an angle of 30° to the horizontal.
- a Determine the vertical component of the velocity and use this value to find the time between leaving the ground and landing. [2]
 - b Determine the horizontal component of the velocity and use this value to find the horizontal distance travelled. [4]
- 6 Figure 2.34 shows an arrangement used to measure the acceleration of a metal plate as it falls vertically. The metal plate is released from rest and falls a distance of 0.200 m before breaking light beam 1. It then falls a further 0.250 m before breaking light beam 2.

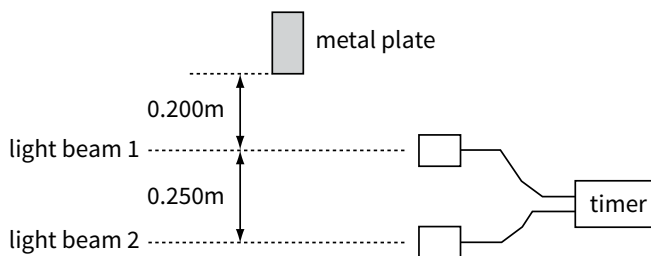


Figure 2.34 For End-of-chapter Question 6.

- a Calculate the time taken for the plate to fall 0.200 m from rest. (You may assume that the metal plate falls with an acceleration equal to the acceleration of free fall.) [2]
- b The timer measures the speed of the metal plate as it falls through each light beam. The speed as it falls through light beam 1 is 1.92 m s^{-1} and the speed as it falls through light beam 2 is 2.91 m s^{-1} .
 - i Calculate the acceleration of the plate between the two light beams. [2]
 - ii State and explain **one** reason why the acceleration of the plate is not equal to the acceleration of free fall. [2]

- 7 Figure 2.35 shows the velocity–time graph for a vertically bouncing ball. The ball is released at A and strikes the ground at B. The ball leaves the ground at D and reaches its maximum height at E. The effects of air resistance can be neglected.

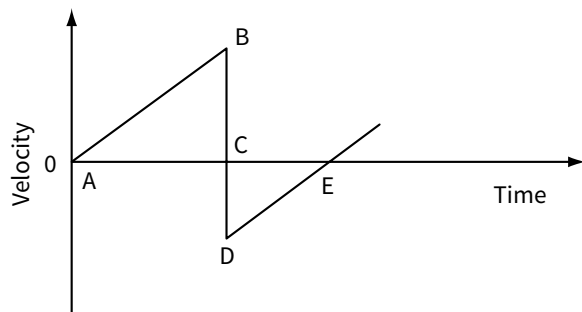


Figure 2.35 For End-of-chapter Question 7.

- a State:
- why the velocity at D is negative [1]
 - why the gradient of the line AB is the same as the gradient of line DE [1]
 - what is represented by the area between the line AB and the time axis [1]
 - why the area of triangle ABC is greater than the area of triangle CDE. [1]
- b The ball is dropped from rest from an initial height of 1.2 m. After hitting the ground the ball rebounds to a height of 0.80 m. The ball is in contact with the ground between B and D for a time of 0.16 s. Using the acceleration of free fall, calculate:
- the speed of the ball immediately before hitting the ground [2]
 - the speed of the ball immediately after hitting the ground [2]
 - the acceleration of the ball while it is in contact with the ground. State the direction of this acceleration. [3]

- 8 A student measures the speed v of a trolley as it moves down a slope. The variation of v with time t is shown in the graph in Figure 2.36.

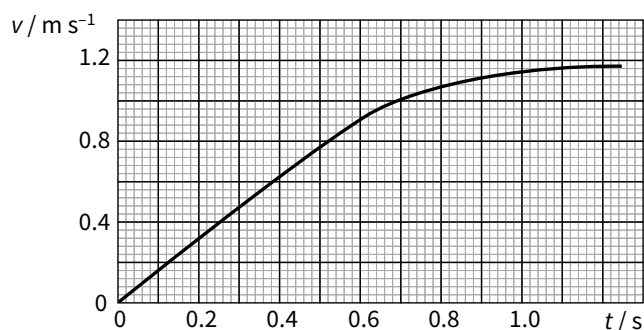


Figure 2.36 For End-of-chapter Question 8.

- Use the graph to find the acceleration of the trolley when $t = 0.7$ s. [2]
- State how the acceleration of the trolley varies between $t = 0$ and $t = 1.0$ s. Explain your answer by reference to the graph. [3]
- Determine the distance travelled by the trolley between $t = 0.6$ and $t = 0.8$ s. [3]
- The student obtained the readings for v using a motion sensor. The readings may have random errors and systematic errors. Explain how these two types of error affect the velocity–time graph. [2]

- 9 A car driver is travelling at speed v on a straight road. He comes over the top of a hill to find a fallen tree on the road ahead. He immediately brakes hard but travels a distance of 60 m at speed v before the brakes are applied. The skid marks left on the road by the wheels of the car are of length 140 m (Figure 2.37). The police investigate whether the driver was speeding and establish that the car decelerates at 2.0 m s^{-2} during the skid.

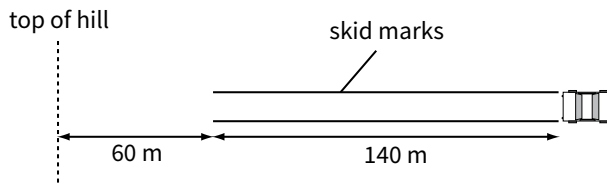


Figure 2.37 For End-of-chapter Question 9.

- Determine the initial speed v of the car before the brakes are applied. [2]
 - Determine the time taken between the driver coming over the top of the hill and applying the brakes. Suggest whether this shows whether the driver was alert to the danger. [2]
 - The speed limit on the road is 100 km/h. Determine whether the driver was breaking the speed limit. [2]
- 10 A hot-air balloon rises vertically. At time $t = 0$, a ball is released from the balloon. Figure 2.38 shows the variation of the ball's velocity v with t . The ball hits the ground at $t = 4.1 \text{ s}$.

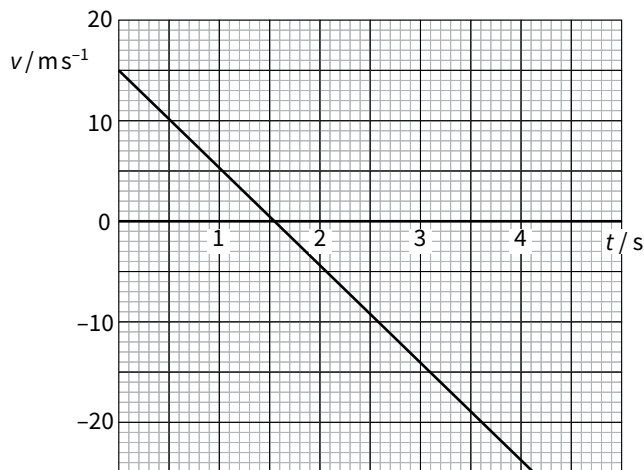


Figure 2.38 For End-of-chapter Question 10.

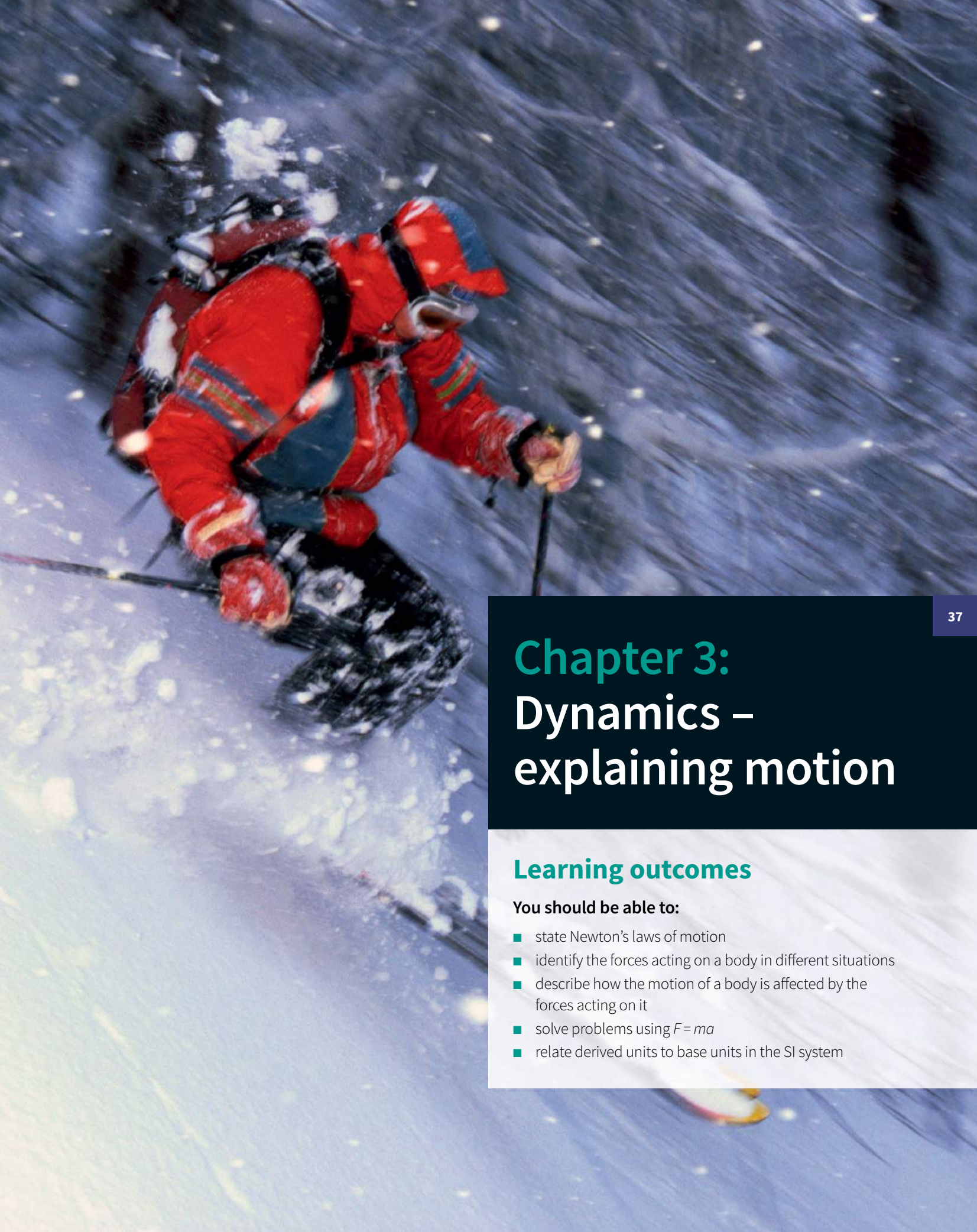
- Explain how the graph shows that the acceleration of the ball is constant. [1]
- Use the graph to:
 - determine the time at which the ball reaches its highest point [1]
 - show that the ball rises for a further 12 m between release and its highest point [2]
 - determine the distance between the highest point reached by the ball and the ground. [2]
- The equation relating v and t is $v = 15 - 9.81t$. Explain the significance in the equation of:
 - the number 15 [1]
 - the negative sign. [1]

- 11** An aeroplane is travelling horizontally at a speed of 80 m s^{-1} and drops a crate of emergency supplies (Figure 2.39). To avoid damage, the maximum vertical speed of the crate on landing is 20 m s^{-1} . You may assume air resistance is negligible.



Figure 2.39 For End-of-chapter Question 11.

- a** Calculate the maximum height of the aeroplane when the crate is dropped. [2]
- b** Calculate the time taken for the crate to reach the ground from this height. [2]
- c** The aeroplane is travelling at the maximum permitted height. Calculate the horizontal distance travelled by the crate after it is released from the aeroplane. [1]



Chapter 3: Dynamics – explaining motion

Learning outcomes

You should be able to:

- state Newton's laws of motion
- identify the forces acting on a body in different situations
- describe how the motion of a body is affected by the forces acting on it
- solve problems using $F = ma$
- relate derived units to base units in the SI system

Force and acceleration

If you have ever flown in an aeroplane you will know how the back of the seat pushes you forwards when the aeroplane accelerates down the runway (Figure 3.1). The pilot must control many forces on the aeroplane to ensure a successful take-off.

In Chapters 1 and 2 we saw how motion can be **described** in terms of displacement, velocity, acceleration and so on. This is known as **kinematics**. Now we are going to look at how we can **explain** how an object moves in terms of the forces which change its motion. This is known as **dynamics**.



Figure 3.1 An aircraft takes off – the force provided by the engines causes the aircraft to accelerate.

Calculating the acceleration

Figure 3.2a shows how we represent the force which the motors on a train provide to cause it to accelerate. The resultant force is represented by a green arrow. The direction of the arrow shows the direction of the resultant force. The magnitude (size) of the resultant force of 20 000 N is also shown.

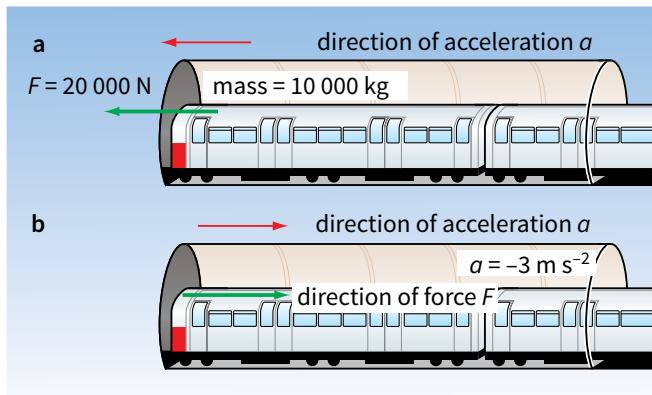


Figure 3.2 A force is needed to make the train **a** accelerate, and **b** decelerate.

To calculate the acceleration a of the train produced by the resultant force F , we must also know the train’s mass m (Table 3.1). These quantities are related by:

$$a = \frac{F}{m} \quad \text{or} \quad F = ma$$

Quantity	Symbol	Unit
resultant force	F	N (newtons)
mass	m	kg (kilograms)
acceleration	a	m s^{-2} (metres per second squared)

Table 3.1 The quantities related by $F = ma$.

In this example we have $F = 20\,000\text{ N}$ and $m = 10\,000\text{ kg}$, and so:

$$a = \frac{F}{m} = \frac{20\,000}{10\,000} = 2\text{ m s}^{-2}$$

In Figure 3.2b, the train is decelerating as it comes into a station. Its acceleration is -3.0 m s^{-2} . What force must be provided by the braking system of the train?

$$F = ma = 10\,000 \times -3 = -30\,000\text{ N}$$

The minus sign shows that the force must act towards the right in the diagram, in the opposite direction to the motion of the train.

Force, mass and acceleration

The equation we used above, $F = ma$, is a simplified version of **Newton’s second law** of motion.

For a body of constant mass, its acceleration is directly proportional to the resultant force applied to it.

An alternative form of Newton’s second law is given in Chapter 6 when you have studied momentum. Since Newton’s second law holds for objects that have a constant mass, this equation can be applied to a train whose mass remains constant during its journey. The equation $a = \frac{F}{m}$ relates acceleration, resultant force and mass. In particular, it shows that the bigger the force, the greater the acceleration it produces. You will probably feel that this is an unsurprising result. For a given object, the acceleration is directly proportional to the resultant force:

$$a \propto F$$